MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT SUBSTATION

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2010

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	26-10-2023	AS BUILT	SHD	SGC	PRI	
C02	26-11-2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	SHD	SGC	PRI	
C01	04-11-2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	SHD	SGC	PRI	
A01	12-10-2020	FIRST ISSUE	SHD	SGC	PRI	

PROPRIETARY & CONFIDENTIAL This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

Ma'aden Document Number:

MD-216-49PR-EG-CV-CAL-2010 D01

Outotec Document Number:

900707 GAU01 WG01 CED01 45019 OU100688144 3

L&T Document Number:

O19045 008 C K 24293 003 3

PROPRIETARY & CONFIDENTIAL : The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

PROPRIETARY & CONFIDENTIAL : This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

TABLE OF CONTENTS

		PAGE NO
DEFINITIONS and ABBREVIATIONS		3
1	INTENT	4
2	EXECUTIVE SUMMARY	4
3	SCOPE	4
4	DESIGN OF POWER PLANT SUBSTATION COLUMN FOUNDATION	5
4.1	Design of Foundation mkd. F1	5
4.2	Design of Foundation mkd. F2	11
4.3	Design of Foundation mkd. F3	14
4.4	Design of Foundation mkd. F4	19
4.5	Design of RC Frame mkd. FR-1	22
REFERENCES		27
ATTACHMENT 1	FOUNDATION LOAD DATA FOR SUBSTATION COLUMNS	28
ATTACHMENT 2	STAAD INPUT & OUTPUT FILE FOR FOUNDATION MARKED F2	32
ATTACHMENT 3	STAAD INPUT & OUTPUT FILE FOR FOUNDATION MARKED F4	47
ATTACHMENT 4	STAAD INPUT & OUTPUT FILE FOR R C FRAME MARKED FR-1	62

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023			Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI		

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
Employer	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works		Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

1. INTENT

The document is prepared for Design Calculation of Column Foundation for **Power Plant Substation.**

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The document is prepared for Design Calculation of Column Foundation for Power Plant Substation.
All loads applied to the Foundation followed the load data provided by Siemens for Substation Building.
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008
Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

4 DESIGN OF POWER PLANT SUBSTATION COLUMN FOUNDATION

4.1 Design of Foundation mkd. F1

Load Detail	Col Mkd. N3A		
	FX (T)	FY (T)	FZ (T)
1 DL	0.0	4.6	1.0
2 LL	0.0	7.0	1.0
3 WL1(+X)	0.0	0.0	0.0
4 WL2(-X)	0.0	0.0	0.0
5 WL3(+Z)	0.0	-0.1	-0.4
6 WL4(-Z)	0.0	0.1	0.1
7 SL1(+X)	-0.1	-0.1	0.0
8 SL2(-X)	0.1	0.1	0.0
9 SL3(+Z)	0.1	0.1	0.0
10 SL4(-Z)	-0.1	0.1	0.2

(Foundation has been designed for the most critical support reactions)

(Refer Foundation Detail dwg no. MD-216-49PR-EG-CV-GAD-2011 & Foundation Load Data)

Calculation of Platform load

Dead Load Intensity = 0.10 T/M^3

Live Load Intensity = 1.50 T/m^2

Influence area of platform = (4.45/2+1.425) * 2.5 m^2

9.13 m^2

Dead load on column = 0.91 T

Live load on column = 13.69 T

Load due to platfrom has been added to load from Foundation load Data for Column Mkd, N1A, N1B,N7A, N7B-N12B, N3C & N6C

But Foundation Load Data for Column mkd. N3A has been found most critical, Hence Foundation has been designed for col. Mkd. N3A & applied for all columns below Pre-Fabricated Electrical Room

Strength Load Combinations as per Cl. No. 2.3.2. SBC 301:2007 for the most critical support reactions as follows:

STRENGTH DESIGN	Col Mkd. B3		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	0.0	6.44	1.40
1.4D+1.7L	0.0	18.3	3.1
1.2D + 1.6WL1+0.5*L	0.0	9.0	1.7
1.2D + 1.6WL2+0.5*L	0.0	9.0	1.7
1.2D + 1.6WL3+0.5*L	0.0	8.9	1.1
1.2D + 1.6WL4+0.5*L	0.0	9.2	1.9
1.2D + 1.0EL1+0.5*L	-0.1	9.0	1.7
1.2D + 1.0EL2+0.5*L	0.1	9.1	1.7
1.2D + 1.0EL3+0.5*L	0.1	9.1	1.7
1.2D + 1.0EL4+0.5*L	-0.1	9.1	1.9
0.9D+1.6WL1	0.0	4.1	0.9
0.9D+1.6WL2	0.0	4.1	0.9
0.9D+1.6WL3	0.0	4.0	0.3
0.9D+1.6WL4	0.0	4.3	1.1
0.9D+1.0EL1	-0.1	4.1	0.9
0.9D+1.0EL2	0.1	4.2	0.9
0.9D+1.0EL3	0.1	4.2	0.9
0.9D+1.0EL4	-0.1	4.2	1.1

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

DESIGN OF FOUNDATION MKD. F1

	ABOUT-X	ABOUT-Z
Size of column :	0.65m	0.65m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

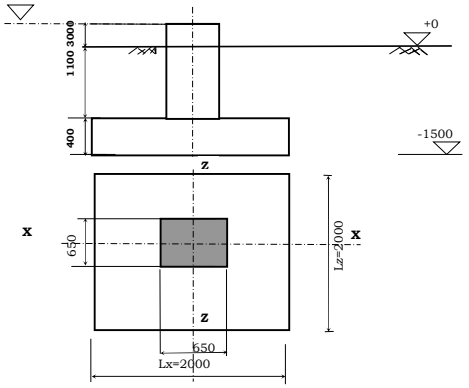
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Monorail Load	MR

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.4*2.5)+(0.65*0.65*4.1*2.5) = 8.33T

Weight of soil = ((2*2)-(0.65*0.65))*1.1*1.8 = 7.08T

U/S Base Pl Total = 15.41T



Design Input:

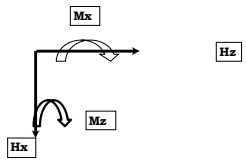
Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 319mm = 0.319m

Effective depth of slab along Z direction = 305mm = 0.305m

Clear projection along Z direction = 0.675m

Clear projection along X direction = 0.675m



WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
D+L	0.00	11.60	2.00	0.0	0.00	27.01	0.00	9.00	2.00	0.00
DL+0.75*L+0.75*WL1	0.00	9.85	1.75	0.0	0.00	25.26	0.00	7.88	1.75	0.00
DL+0.75*L+0.75*WL2	0.00	9.85	1.75	0.0	0.00	25.26	0.00	7.88	1.75	0.00
DL+0.75*L+0.75*WL3	0.00	9.78	1.45	0.0	0.00	25.19	0.00	6.53	1.45	0.00
DL+0.75*L+0.75*WL4	0.00	9.93	1.83	0.0	0.00	25.34	0.00	8.21	1.83	0.00
DL+0.75*L+0.75*0.7*EL1	-0.05	9.82	1.75	0.0	0.00	25.23	-0.05	7.88	1.75	0.24
DL+0.75*L+0.75*0.7*EL2	0.05	9.90	1.75	0.0	0.00	25.31	0.05	7.88	1.75	0.24
DL+0.75*L+0.75*0.7*EL3	0.05	9.90	1.76	0.0	0.00	25.31	0.05	7.90	1.76	0.24
DL+0.75*L+0.75*0.7*EL4	-0.05	9.90	1.86	0.0	0.00	25.31	-0.05	8.35	1.86	0.24
0.6*DL+WL1	0.00	2.76	0.60	0.0	0.00	18.17	0.00	2.70	0.60	0.00
0.6*DL+WL2	0.00	2.76	0.60	0.0	0.00	18.17	0.00	2.70	0.60	0.00
0.6*DL+WL3	0.00	2.66	0.20	0.0	0.00	18.07	0.00	0.90	0.20	0.00
0.6*DL+WL4	0.00	2.86	0.70	0.0	0.00	18.27	0.00	3.15	0.70	0.00
0.6*DL+0.7*EL1	-0.07	2.73	0.60	0.0	0.00	18.14	-0.07	2.70	0.60	0.32
0.6*DL+0.7*EL2	0.07	2.83	0.60	0.0	0.00	18.24	0.07	2.70	0.60	0.32
0.6*DL+0.7*EL3	0.07	2.83	0.61	0.0	0.00	18.24	0.07	2.73	0.61	0.32
0.6*DL+0.7*EL4	-0.07	2.83	0.74	0.0	0.00	18.24	-0.07	3.33	0.74	0.32

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
D+L	27.01	0.00	0.00	2.00	9.00	-	3.00	-	5.40
DL+0.75*L+0.75*WL1	25.26	0.00	0.00	1.75	7.88	-	3.21	-	5.77
DL+0.75*L+0.75*WL2	25.26	0.00	0.00	1.75	7.88	-	3.21	-	5.77
DL+0.75*L+0.75*WL3	25.19	0.00	0.00	1.45	6.53	-	3.86	-	6.95
DL+0.75*L+0.75*WL4	25.34	0.00	0.00	1.83	8.21	-	3.08	-	5.55
DL+0.75*L+0.75*0.7*EL1	25.23	-0.05	0.24	1.75	7.88	106.81	3.20	192.26	5.77
DL+0.75*L+0.75*0.7*EL2	25.31	0.05	0.24	1.75	7.88	107.14	3.21	192.86	5.79
DL+0.75*L+0.75*0.7*EL3	25.31	0.05	0.24	1.76	7.90	107.14	3.20	192.86	5.77
DL+0.75*L+0.75*0.7*EL4	25.31	-0.05	0.24	1.86	8.35	107.14	3.03	192.86	5.46
0.6*DL+WL1	18.17	0.00	0.00	0.60	2.70	-	6.73	-	12.11
0.6*DL+WL2	18.17	0.00	0.00	0.60	2.70	-	6.73	-	12.11
0.6*DL+WL3	18.07	0.00	0.00	0.20	0.90	-	20.08	-	36.14
0.6*DL+WL4	18.27	0.00	0.00	0.70	3.15	-	5.80	-	10.44
0.6*DL+0.7*EL1	18.14	-0.07	0.32	0.60	2.70	57.57	6.72	103.63	12.09
0.6*DL+0.7*EL2	18.24	0.07	0.32	0.60	2.70	57.90	6.76	104.23	12.16
0.6*DL+0.7*EL3	18.24	0.07	0.32	0.61	2.73	57.90	6.68	104.23	12.02
0.6*DL+0.7*EL4	18.24	-0.07	0.32	0.74	3.33	57.90	5.48	104.23	9.86

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	0.00	6.44	1.40	0.0	0.00	21.850	6.30	0.00
1.4D+1.7L	0.00	18.34	3.10	0.0	0.00	33.750	13.95	0.00
1.2D + 1.6WL1+0.5*L	0.00	9.02	1.70	0.0	0.00	24.430	7.65	0.00
1.2D + 1.6WL2+0.5*L	0.00	9.02	1.70	0.0	0.00	24.430	7.65	0.00
1.2D + 1.6WL3+0.5*L	0.00	8.86	1.06	0.0	0.00	24.270	4.77	0.00
1.2D + 1.6WL4+0.5*L	0.00	9.18	1.86	0.0	0.00	24.590	8.37	0.00
1.2D + 1.0EL1+0.5*L	-0.10	8.97	1.70	0.0	0.00	24.380	7.65	0.45
1.2D + 1.0EL2+0.5*L	0.10	9.12	1.70	0.0	0.00	24.530	7.65	0.45
1.2D + 1.0EL3+0.5*L	0.10	9.12	1.71	0.0	0.00	24.530	7.70	0.45
1.2D + 1.0EL4+0.5*L	-0.10	9.12	1.90	0.0	0.00	24.530	8.55	0.45
0.9D+1.6WL1	0.00	4.14	0.90	0.0	0.00	19.550	4.05	0.00
0.9D+1.6WL2	0.00	4.14	0.90	0.0	0.00	19.550	4.05	0.00
0.9D+1.6WL3	0.00	3.98	0.26	0.0	0.00	19.390	1.17	0.00
0.9D+1.6WL4	0.00	4.30	1.06	0.0	0.00	19.710	4.77	0.00
0.9D+1.0EL1	-0.10	4.09	0.90	0.0	0.00	19.500	4.05	0.45
0.9D+1.0EL2	0.10	4.24	0.90	0.0	0.00	19.650	4.05	0.45
0.9D+1.0EL3	0.10	4.24	0.91	0.0	0.00	19.650	4.10	0.45
0.9D+1.0EL4	-0.10	4.24	1.10	0.0	0.00	19.650	4.95	0.45

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

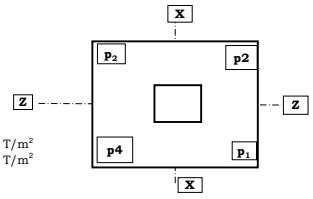
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² (BH28) For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 58.1 T/m² For WL/SL case = 70.6 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m²) V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-ph)	Remarks
D+L	27.01	0.00	9.00	6.75	0.00	6.75	15.50	2.00	15.50	2.00	12.80	O.K
DL+0.75*L+0.75*WL1	25.26	0.00	7.88	6.32	0.00	5.91	14.22	2.41	14.22	2.41	11.52	O.K
DL+0.75*L+0.75*WL2	25.26	0.00	7.88	6.32	0.00	5.91	14.22	2.41	14.22	2.41	11.52	O.K
DL+0.75*L+0.75*WL3	25.19	0.00	6.53	6.30	0.00	4.89	13.19	3.40	13.19	3.40	10.49	O.K
DL+0.75*L+0.75*WL4	25.34	0.00	8.21	6.33	0.00	6.16	14.49	2.17	14.49	2.17	11.79	O.K
DL+0.75*L+0.75*0.7*EL1	25.23	0.24	7.88	6.31	0.18	5.91	14.39	2.58	14.04	2.23	11.69	O.K
DL+0.75*L+0.75*0.7*EL2	25.31	0.24	7.88	6.33	0.18	5.91	14.41	2.60	14.06	2.24	11.71	O.K
DL+0.75*L+0.75*0.7*EL3	25.31	0.24	7.90	6.33	0.18	5.92	14.43	2.58	14.07	2.23	11.73	O.K
DL+0.75*L+0.75*0.7*EL4	25.31	0.24	8.35	6.33	0.18	6.26	14.77	2.24	14.41	1.89	12.07	O.K
0.6*DL+WL1	18.17	0.00	2.70	4.54	0.00	2.03	8.57	4.52	8.57	4.52	5.87	O.K
0.6*DL+WL2	18.17	0.00	2.70	4.54	0.00	2.03	8.57	4.52	8.57	4.52	5.87	O.K
0.6*DL+WL3	18.07	0.00	0.90	4.52	0.00	0.68	7.19	5.84	7.19	5.84	4.49	O.K
0.6*DL+WL4	18.27	0.00	3.15	4.57	0.00	2.36	8.93	4.21	8.93	4.21	6.23	O.K
0.6*DL+0.7*EL1	18.14	0.32	2.70	4.53	0.24	2.03	8.80	4.75	8.32	4.27	6.10	O.K
0.6*DL+0.7*EL2	18.24	0.32	2.70	4.56	0.24	2.03	8.82	4.77	8.35	4.30	6.12	O.K
0.6*DL+0.7*EL3	18.24	0.32	2.73	4.56	0.24	2.05	8.84	4.75	8.37	4.28	6.14	O.K
0.6*DL+0.7*EL4	18.24	0.32	3.33	4.56	0.24	2.50	9.29	4.30	8.82	3.83	6.59	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

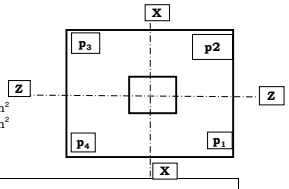
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 58.1 T/m² For WL/SL case = 70.6 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-ph)
1.4D	21.9	0.00	6.30	5.46	0.00	4.73	12.19	2.74	12.19	2.74	9.49
1.4D+1.7L	33.8	0.00	13.95	8.44	0.00	10.46	20.90	-2.03	18.90	-2.03	18.92
1.2D + 1.6WL1+0.5*L	24.4	0.00	7.65	6.11	0.00	5.74	13.85	0.37	11.85	0.37	11.87
1.2D + 1.6WL2+0.5*L	24.4	0.00	7.65	6.11	0.00	5.74	13.85	0.37	11.85	0.37	11.87
1.2D + 1.6WL3+0.5*L	24.3	0.00	4.77	6.07	0.00	3.58	11.65	2.49	9.65	2.49	9.67
1.2D + 1.6WL4+0.5*L	24.6	0.00	8.37	6.15	0.00	6.28	14.43	-0.13	12.43	-0.13	12.45
1.2D + 1.0EL1+0.5*L	24.4	0.45	7.65	6.10	0.34	5.74	14.17	0.70	11.50	0.02	12.19
1.2D + 1.0EL2+0.5*L	24.5	0.45	7.65	6.13	0.34	5.74	14.21	0.73	11.53	0.06	12.23
1.2D + 1.0EL3+0.5*L	24.5	0.45	7.70	6.13	0.34	5.77	14.24	0.70	11.57	0.02	12.26
1.2D + 1.0EL4+0.5*L	24.5	0.45	8.55	6.13	0.34	6.41	14.88	0.06	12.21	-0.62	12.90
0.9D+1.6WL1	19.6	0.00	4.05	4.89	0.00	3.04	9.93	1.85	7.93	1.85	7.95
0.9D+1.6WL2	19.6	0.00	4.05	4.89	0.00	3.04	9.93	1.85	7.93	1.85	7.95
0.9D+1.6WL3	19.4	0.00	1.17	4.85	0.00	0.88	7.73	3.97	5.73	3.97	5.75
0.9D+1.6WL4	19.7	0.00	4.77	4.93	0.00	3.58	10.51	1.35	8.51	1.35	8.53
0.9D+1.0EL1	19.5	0.45	4.05	4.88	0.34	3.04	10.25	2.18	7.58	1.50	8.27
0.9D+1.0EL2	19.7	0.45	4.05	4.91	0.34	3.04	10.29	2.21	7.61	1.54	8.31
0.9D+1.0EL3	19.7	0.45	4.10	4.91	0.34	3.07	10.32	2.18	7.65	1.50	8.34
0.9D+1.0EL4	19.7	0.45	4.95	4.91	0.34	3.71	10.96	1.54	8.29	0.86	8.98

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.65+2*0.5*0.305)+(0.65+2*0.5*0.305))*2

= 3.82 m

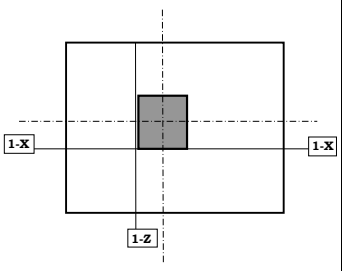
Punching Area= ((0.65+2*0.5*0.305)*(0.65+2*0.5*0.305))

= 0.91 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*2.5+1.1*1.8)

= 2.98 T/m^2

Area of footing, A_f= 4.00 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 58.46 T

Strength of Concrete in shear Vc1 = 249.18 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 298.33 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 229.76 T **Table 22.6.5.2_ (c)**

Design Vc= 229.76 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (58.46*10000)/(3820*305)

= 0.50 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	9.49	2.16	3.38	2.16	3.51	29.32	0.25	OK	0.74	0.115	OK	0.23
1.4D+1.7L	18.92	4.31	6.74	4.31	7.00	58.46	0.5	OK	0.74	0.23	OK	0.46
1.2D + 1.6WL1+0.5*L	11.87	2.70	4.23	2.70	4.39	36.68	0.31	OK	0.74	0.144	OK	0.29
1.2D + 1.6WL2+0.5*L	11.87	2.70	4.23	2.70	4.39	36.68	0.31	OK	0.74	0.144	OK	0.29
1.2D + 1.6WL3+0.5*L	9.67	2.20	3.44	2.20	3.58	29.88	0.26	OK	0.74	0.117	OK	0.24
1.2D + 1.6WL4+0.5*L	12.45	2.84	4.43	2.84	4.61	38.47	0.33	OK	0.74	0.151	OK	0.30
1.2D + 1.0EL1+0.5*L	12.19	2.78	4.34	2.78	4.51	37.67	0.32	OK	0.74	0.148	OK	0.30
1.2D + 1.0EL2+0.5*L	12.23	2.79	4.35	2.79	4.53	37.79	0.32	OK	0.74	0.148	OK	0.30
1.2D + 1.0EL3+0.5*L	12.26	2.79	4.36	2.79	4.54	37.88	0.33	OK	0.74	0.149	OK	0.30
1.2D + 1.0EL4+0.5*L	12.90	2.94	4.59	2.94	4.77	39.86	0.34	OK	0.74	0.156	OK	0.32
0.9D+1.6WL1	7.95	1.81	2.83	1.81	2.94	24.57	0.21	OK	0.74	0.096	OK	0.19
0.9D+1.6WL2	7.95	1.81	2.83	1.81	2.94	24.57	0.21	OK	0.74	0.096	OK	0.19
0.9D+1.6WL3	5.75	1.31	2.05	1.31	2.13	17.77	0.15	OK	0.74	0.07	OK	0.14
0.9D+1.6WL4	8.53	1.94	3.04	1.94	3.16	26.36	0.23	OK	0.74	0.103	OK	0.21
0.9D+1.0EL1	8.27	1.88	2.94	1.88	3.06	25.55	0.22	OK	0.74	0.1	OK	0.20
0.9D+1.0EL2	8.31	1.89	2.96	1.89	3.07	25.68	0.22	OK	0.74	0.101	OK	0.20
0.9D+1.0EL3	8.34	1.90	2.97	1.90	3.09	25.77	0.22	OK	0.74	0.101	OK	0.20
0.9D+1.0EL4	8.98	2.05	3.20	2.05	3.32	27.75	0.24	OK	0.74	0.109	OK	0.22

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.31	0.001	319
1.4D+1.7L	0.62	0.001	319
1.2D + 1.6WL1+0.5*L	0.39	0.001	319
1.2D + 1.6WL2+0.5*L	0.39	0.001	319
1.2D + 1.6WL3+0.5*L	0.32	0.001	319
1.2D + 1.6WL4+0.5*L	0.41	0.001	319
1.2D + 1.0EL1+0.5*L	0.4	0.001	319
1.2D + 1.0EL2+0.5*L	0.4	0.001	319
1.2D + 1.0EL3+0.5*L	0.4	0.001	319
1.2D + 1.0EL4+0.5*L	0.42	0.001	319
0.9D+1.6WL1	0.26	0.001	319
0.9D+1.6WL2	0.26	0.001	319
0.9D+1.6WL3	0.19	0	0
0.9D+1.6WL4	0.28	0.001	319
0.9D+1.0EL1	0.27	0.001	319
0.9D+1.0EL2	0.27	0.001	319
0.9D+1.0EL3	0.27	0.001	319
0.9D+1.0EL4	0.29	0.001	319

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd. (mm)
At bottom	4.31	0.46	0.62	0.001	319.00	574.20	12ϕT	196.97	300	150
At top	0.00	0.00	0.00	0.0018	549.00	549.00	12ϕT	206.01	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**150**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**150**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

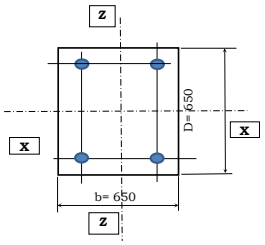
Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTIO N & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

DESIGN OF COLUMN MKD P1

Input:

Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1700

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) = 4500.00
Unsupported length of column about Z (l_{uz}) = 4500.00

For Columns not braced against sidesway,
Slenderness ratio, $K\,l_{ux} / r_x$ = 19.19 < 22 SECTION 6.2.5a
$K\,l_{uz} / r_z$ = 19.19 < 22 SECTION 6.2.5a

Where, Effective length factor K= 0.8
Radius of gyration r_x= 188
Radius of gyration r_z= 188

γ_y = 0.73
γ_z = 0.73

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = $P_n/f_c'bD$ (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	0.00	6.44	1.40	0.0	9.9	3.7	0.01
1.4D+1.7L	0.00	18.34	3.10	0.0	28.2	8.1	0.02
1.2D + 1.6WL1+0.5*L	0.00	9.02	1.70	0.0	13.9	4.4	0.01
1.2D + 1.6WL2+0.5*L	0.00	9.02	1.70	0.0	13.9	4.4	0.01
1.2D + 1.6WL3+0.5*L	0.00	8.86	1.06	0.0	13.6	2.8	0.01
1.2D + 1.6WL4+0.5*L	0.00	9.18	1.86	0.0	14.1	4.9	0.01
1.2D + 1.0EL1+0.5*L	-0.10	8.97	1.70	0.3	13.8	4.4	0.01
1.2D + 1.0EL2+0.5*L	0.10	9.12	1.70	0.3	14.0	4.4	0.01
1.2D + 1.0EL3+0.5*L	0.10	9.12	1.71	0.3	14.0	4.5	0.01
1.2D + 1.0EL4+0.5*L	-0.10	9.12	1.90	0.3	14.0	5.0	0.01
0.9D+1.6WL1	0.00	4.14	0.90	0.0	6.4	2.4	0.00
0.9D+1.6WL2	0.00	4.14	0.90	0.0	6.4	2.4	0.00
0.9D+1.6WL3	0.00	3.98	0.26	0.0	6.1	0.7	0.00
0.9D+1.6WL4	0.00	4.30	1.06	0.0	6.6	2.8	0.00
0.9D+1.0EL1	-0.10	4.09	0.90	0.3	6.3	2.4	0.00
0.9D+1.0EL2	0.10	4.24	0.90	0.3	6.5	2.4	0.00
0.9D+1.0EL3	0.10	4.24	0.91	0.3	6.5	2.4	0.00
0.9D+1.0EL4	-0.10	4.24	1.10	0.3	6.5	2.9	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. = 4479 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm	(At middle)	
4-Nos	20 mm Dia	1257.14 Sq mm	(At middle)	Total A_{sc} = 1.1 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85\,f_c'(A_g - A_{sc}) + f_y A_{sc}$ = 1431.7138 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_c'b^2D$	$R_{nx}=M_{nox}/f_c'bD^2$						
1.4D	0.05	0.05	48.1	48.06	0.55	1.16	0.05	O.K
1.4D+1.7L	0.05	0.05	48.1	48.06	0.55	1.16	0.13	O.K
1.2D + 1.6WL1+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.06	O.K
1.2D + 1.6WL2+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.06	O.K
1.2D + 1.6WL3+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.04	O.K
1.2D + 1.6WL4+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.07	O.K
1.2D + 1.0EL1+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.07	O.K
1.2D + 1.0EL2+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.07	O.K
1.2D + 1.0EL3+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.07	O.K
1.2D + 1.0EL4+0.5*L	0.05	0.05	48.1	48.06	0.55	1.16	0.07	O.K
0.9D+1.6WL1	0.05	0.05	48.1	48.06	0.55	1.16	0.03	O.K
0.9D+1.6WL2	0.05	0.05	48.1	48.06	0.55	1.16	0.03	O.K
0.9D+1.6WL3	0.05	0.05	48.1	48.06	0.55	1.16	0.01	O.K
0.9D+1.6WL4	0.05	0.05	48.1	48.06	0.55	1.16	0.04	O.K
0.9D+1.0EL1	0.05	0.05	48.1	48.06	0.55	1.16	0.03	O.K
0.9D+1.0EL2	0.05	0.05	48.1	48.06	0.55	1.16	0.03	O.K
0.9D+1.0EL3	0.05	0.05	48.1	48.06	0.55	1.16	0.03	O.K
0.9D+1.0EL4	0.05	0.05	48.1	48.06	0.55	1.16	0.04	O.K

Links, **10mm dia @ 200 C/C**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

4.2 Design of Foundation mkd. F2 & F2A (Refer Load Data For NODE MARKED N1E & N2E)

	N1E			N2E		
LOAD ITEM	Fx (T)	Fy (T)	Fz (T)	Fx (T)	Fy (T)	Fz (T)
1 DL	0.00	1.50	-0.06	0.00	1.50	-0.06
2 LL	0.00	12.50	-0.06	0.00	12.50	-0.06
3 WL1(+X)	0.00	-0.01	0.00	0.00	0.01	0.00
4 WL2(-X)	0.00	0.01	0.00	0.00	-0.01	0.00
5 WL3(+Z)	0.00	0.06	-0.11	0.00	0.07	-0.10
6 WL4(-Z)	0.00	-0.06	0.11	0.00	-0.07	0.10
7 SL1(+X)	-0.02	-0.05	0.00	-0.02	0.06	-0.01
8 SL2(-X)	0.02	0.05	0.00	0.02	-0.06	0.01
9 SL3(+Z)	0.02	0.05	0.00	0.02	-0.06	0.01
10 SL4(-Z)	0.00	-0.06	0.02	0.00	-0.06	0.02

Calculation of Platform load

Dead Load Intensity =	0.1	T/m^2
Dead Load Intensity =		T/M^3
Live Load Intensity =	1	T/m^2
Total Load =	1.1	T/m^2
Influence area of platform =	2.4 * 3.7	m^2
	8.9	m^2
Dead load on column =	0.888	T
Live load on column =	8.9	T

Load due to platfrom has been added to Foundation Load Data of each column

FOUNDATION SIZE 1.15m X 3.70m
DEPTH OF THE FOUNDATION 1.50m (below FGL)
THE WT OF SOIL +SELF WT 1.15X3.7X1.5XUNIT WT = 12.765 T

WORKING STRESS DESIGN

	C-1			C-2			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	14.00	-0.11	0.00	14.00	-0.11	40.77	0.00	1.02	-0.23	0.00
DL+0.75*L+0.75*WL1	0.00	10.89	-0.10	-0.01	10.89	-0.10	34.54	0.00	0.89	-0.20	0.03
DL+0.75*L+0.75*WL2	0.01	10.91	-0.10	0.00	10.87	-0.10	34.54	0.01	0.89	-0.20	0.07
DL+0.75*L+0.75*WL3	0.00	10.94	-0.18	-0.01	10.93	-0.18	34.64	-0.01	1.61	-0.36	0.04
DL+0.75*L+0.75*WL4	0.01	10.86	-0.02	0.00	10.82	-0.02	34.45	0.01	0.18	-0.04	0.07
DL+0.75*L+0.75*0.7*EL1	-0.01	10.87	-0.10	-0.01	10.91	-0.10	34.55	-0.02	0.90	-0.20	0.13
DL+0.75*L+0.75*0.7*EL2	0.01	10.93	-0.10	0.01	10.85	-0.10	34.54	0.02	0.88	-0.20	0.19
DL+0.75*L+0.75*0.7*EL3	0.01	10.93	-0.10	0.01	10.85	-0.10	34.54	0.02	0.88	-0.20	0.19
DL+0.75*L+0.75*0.7*EL4	0.00	10.87	-0.09	0.00	10.85	-0.09	34.48	0.00	0.78	-0.17	0.03
0.6*DL+WL1	0.00	0.91	-0.03	-0.01	0.91	-0.03	14.58	-0.01	0.30	-0.07	0.04
0.6*DL+WL2	0.01	0.93	-0.03	0.00	0.89	-0.03	14.58	0.01	0.29	-0.07	0.07
0.6*DL+WL3	0.00	0.97	-0.14	-0.01	0.97	-0.13	14.71	-0.01	1.25	-0.28	0.04
0.6*DL+WL4	0.01	0.86	0.08	0.00	0.83	0.07	14.45	0.01	0.65	0.15	0.07
0.6*DL+0.7*EL1	-0.01	0.9	0.0	-0.02	0.94	-0.04	14.59	-0.03	0.31	-0.07	0.20
0.6*DL+0.7*EL2	0.02	1.0	0.0	0.01	0.86	-0.03	14.58	0.03	0.28	-0.06	0.23
0.6*DL+0.7*EL3	0.02	1.0	0.0	0.01	0.86	-0.03	14.58	0.03	0.28	-0.06	0.23
0.6*DL+0.7*EL4	0.00	0.9	0.0	0.00	0.86	-0.02	14.50	0.00	0.14	-0.03	0.02

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	40.77	0.00	0.00	-0.23	1.02	20948.68	22.94	20382.50	71.80
DL+0.75*L+0.75*WL1	34.54	0.00	0.03	-0.20	0.89	2308.14	22.24	3070.58	69.63
DL+0.75*L+0.75*WL2	34.54	0.01	0.07	-0.20	0.89	917.02	22.36	2631.86	70.00
DL+0.75*L+0.75*WL3	34.64	-0.01	0.04	-0.36	1.61	1606.42	12.41	2399.39	38.85
DL+0.75*L+0.75*WL4	34.45	0.01	0.07	-0.04	0.18	924.14	112.42	2111.63	351.94
DL+0.75*L+0.75*0.7*EL1	34.55	-0.02	0.13	-0.20	0.90	475.83	22.00	715.52	68.88
DL+0.75*L+0.75*0.7*EL2	34.54	0.02	0.19	-0.20	0.88	334.23	22.61	688.67	70.76
DL+0.75*L+0.75*0.7*EL3	34.54	0.02	0.19	-0.20	0.88	334.23	22.61	-	70.76
DL+0.75*L+0.75*0.7*EL4	34.48	0.00	0.03	-0.17	0.78	1907.71	25.56	-	80.00
0.6*DL+WL1	14.58	-0.01	0.04	-0.07	0.30	651.96	27.84	-	87.14
0.6*DL+WL2	14.58	0.01	0.07	-0.07	0.29	393.12	28.43	-	88.99
0.6*DL+WL3	14.71	-0.01	0.04	-0.28	1.25	673.62	6.76	-	21.17
0.6*DL+WL4	14.45	0.01	0.07	0.15	0.65	395.24	12.70	-	39.74
0.6*DL+0.7*EL1	14.59	-0.03	0.20	-0.07	0.31	132.67	26.70	-	83.57
0.6*DL+0.7*EL2	14.58	0.03	0.23	-0.06	0.28	116.95	29.72	-	93.05
0.6*DL+0.7*EL3	14.58	0.03	0.23	-0.06	0.28	116.95	29.72	-	93.05
0.6*DL+0.7*EL4	14.50	0.00	0.02	-0.03	0.14	1293.47	57.50	-	180.00

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 12 T/m^2 IN LC DL+LL . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 117) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

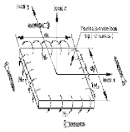
Foundation Thickness =	400 mm
Width =	1000 mm
Clear Cover =	75 mm
Compressive Strength of Concrete , fc' =	35 N/mm^2
Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.832	T-M	Corresponding Mxy	-0.22	T-M	*(Plate No. - 159 , L/C No. - 101)*
Min. Mx	-0.019	T-M	Corresponding Mxy	0.01	T-M	*(Plate No. - 97 , L/C No. - 101)*
Max. My	1.461	T-M	Corresponding Mxy	0.25	T-M	*(Plate No. - 166 , L/C No. - 101)*
Min. My	-7.493	T-M	Corresponding Mxy	0.01	T-M	*(Plate No. - 97 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	1.832 + 0.215 =	2.047 T-M
My1 =	My + abs(Mxy) =	1.461 + 0.25 =	1.711 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	1.832 + 0.032 =	1.864 T-M
My2 =	My+abs(Mxy^2/Mx) =	1.461 + 0.034 =	1.495 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 2.047 T-m
Myd = 1.711 T-m

ALONG X - DIRECTION :

Mu = 2.047 T-m
Φ*Kn = Mu/bd^2 deff. = 307 mm
Mu/bd^2 = 2.047X10^7/(1000X307^2) = 0.217 N/mm^2 < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X307/100 = 1.84 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.711 T-m
Φ*Kn = Mu/bd^2 deff. = 319 mm
Mu/bd^2 = 1.711X10^7/(1000X319^2) = 0.168 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X319/100 = 1.28 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-0.019 - 0.01 =	-0.029 T-M
My1 =	My - abs(Mxy) =	-7.493 - 0.01 =	-7.503 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-0.019 - 0 =	-0.019 T-M
My2 =	My-abs(Mxy^2/Mx) =	-7.493 - 0.005 =	-7.488 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -0.029 T-M
Myd = -7.503 T-M

ALONG X - DIRECTION :

Mu = 0.029 T-m
Φ*Kn = Mu/bd^2 deff. = 307 mm
Mu/bd^2 = 0.029X10^7/(1000X307^2) = 0.003 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X307/100 = 0 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 7.503 T-m
Φ*Kn = Mu/bd^2 deff. = 319 mm
Mu/bd^2 = 7.503X10^7/(1000X319^2) = 0.737 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X319/100 = 6.38 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.021 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 85 , L/C No. - 101)*

Vu = 0.645 T

Provided Pt = 0.0025

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0025X0.645X10^4X307/0.029/10^7)X1000X307/10^4

= 37.969 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0025)X1000X307/10^4

= 30.365 T

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X307/10^4

= 52.671 T

HENCE, Vc = 30.365 T

Φt*Vc = 0.75X30.365 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 22.774 T **> 0.645T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.294 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 85 , L/C No. - 101)*

Vu = 9.379 T

Pt provided = 0.0024

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0024X9.379X10^4X319/7.503/10^7)X1000X319/10^4

= 30.715 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0024)X1000X319/10^4

= 31.497 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X319/10^4

= 54.73 T

HENCE, Vc = 30.715 T

Φt*Vc = 0.75X30.715 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 23.036 T **> 9.379T , Hence O.K.**

PEDESTAL REINFORCEMENT

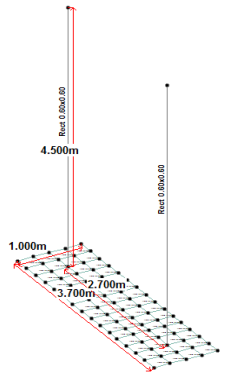
Size of Column = 650 X 650 mm^2

Reqired area of Reinforcement = 4225 mm^2

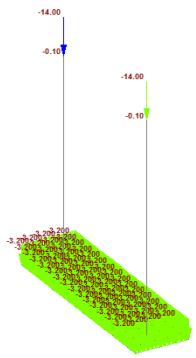
Reinforcement Provided = 8 no. 20 dia bars + 4 no. 25 dia bars

Area of Reinforcement Provided = 4476.8 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2

Soil Weight over the foundation = 1.2x 1.8 T/m2 = 2.16 T/m^2

SOIL+SURCHARGE LOAD = 3.16 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

4.3 Design of Foundation mkd. F3

Load Detail	Col Mkd. 6D'/4D'		
	FX (T)	FY (T)	FZ (T)
1 DL	0.0	0.3	1.0
2 LL	0.0	2.6	1.0
3 WL1(+X)	0.0	0.0	0.0
4 WL2(-X)	0.0	0.0	0.0
5 WL3(+Z)	0.0	0.1	-0.3
6 WL4(-Z)	0.0	-0.1	0.3
7 SL1(+X)	-0.1	0.0	0.0
8 SL2(-X)	0.1	0.0	0.0
9 SL3(+Z)	0.1	0.0	0.0
10 SL4(-Z)	0.0	-0.1	0.1

(Foundation has been designed for the most critical support reactions)

(Refer Foundation Detail dwg no. MD-216-49PR-EG-CV-GAD-2011 & Foundation Load Data)

Dead Load Intensity =	0.1	T/m^2
Live Load Intensity =	1	T/m^2
Total Load =	1.1	T/m^2
Influence area of platform =	1.6 * 1.6	m^2
	2.6	m^2
Dead load on column =	0.256	T
Live load on column =	2.6	T

For Wind & Seismic Load, Refer Foundation Load Data for Column Mkd, N9B & N11B

Strength Load Combinations as per Cl. No. 2.3.2. SBC 301:2007 for the most critical support reactions as follows:

STRENGTH DESIGN	Col Mkd. B3		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	0.0	0.36	1.40
1.4D+1.7L	0.0	4.8	3.1
1.2D + 1.6WL1+0.5*L	0.0	1.6	1.7
1.2D + 1.6WL2+0.5*L	0.0	1.6	1.7
1.2D + 1.6WL3+0.5*L	0.0	1.8	1.3
1.2D + 1.6WL4+0.5*L	0.0	1.5	2.1
1.2D + 1.0EL1+0.5*L	-0.1	1.6	1.7
1.2D + 1.0EL2+0.5*L	0.1	1.6	1.7
1.2D + 1.0EL3+0.5*L	0.1	1.6	1.7
1.2D + 1.0EL4+0.5*L	0.0	1.5	1.8
0.9D+1.6WL1	0.0	0.2	0.9
0.9D+1.6WL2	0.0	0.2	0.9
0.9D+1.6WL3	0.0	0.4	0.5
0.9D+1.6WL4	0.0	0.1	1.3
0.9D+1.0EL1	-0.1	0.2	0.9
0.9D+1.0EL2	0.1	0.2	0.9
0.9D+1.0EL3	0.1	0.2	0.9
0.9D+1.0EL4	0.0	0.2	1.0

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

DESIGN OF FOUNDATION MKD. F3

	ABOUT-X	ABOUT-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.65m	1.65m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Monorail Load	MR

Design Input:

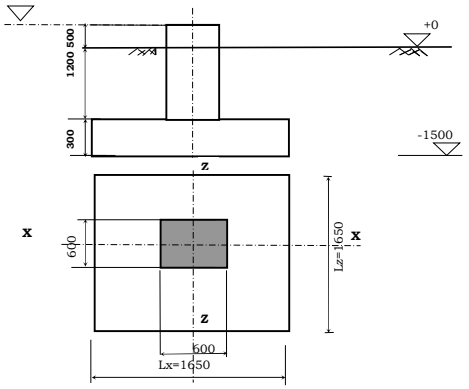
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

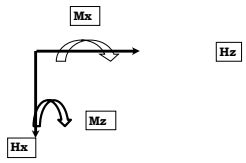
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.65*1.65*0.3*2.5)+(0.6*0.6*1.7*2.5) = 3.57T

Weight of soil = ((1.65*1.65)-(0.6*0.6))*1.2*1.8 = 5.10T

U/S Base Pl Total = 8.67T





Effective depth of slab along X direction = 219mm = 0.219m

Effective depth of slab along Z direction = 205mm = 0.205m

Clear projection along Z direction = 0.525m

Clear projection along X direction = 0.525m

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	2.86	2.00	0.0	0.00	18.27	0.00	4.00	2.00	0.00
DL+0.75*L+0.75*WL1	-0.02	2.21	1.75	0.0	0.00	17.62	-0.02	3.50	1.75	0.05
DL+0.75*L+0.75*WL2	0.02	2.21	1.75	0.0	0.00	17.62	0.02	3.50	1.75	0.05
DL+0.75*L+0.75*WL3	0.00	2.28	1.55	0.0	0.00	17.69	0.00	3.11	1.55	0.00
DL+0.75*L+0.75*WL4	0.00	2.14	1.95	0.0	0.00	17.55	0.00	3.89	1.95	0.00
DL+0.75*L+0.75*0.7*EL1	-0.07	2.20	1.75	0.0	0.00	17.61	-0.07	3.50	1.75	0.15
DL+0.75*L+0.75*0.7*EL2	0.07	2.22	1.75	0.0	0.00	17.63	0.07	3.50	1.75	0.15
DL+0.75*L+0.75*0.7*EL3	0.07	2.22	1.75	0.0	0.00	17.63	0.07	3.50	1.75	0.15
DL+0.75*L+0.75*0.7*EL4	0.00	2.17	1.81	0.0	0.00	17.58	0.00	3.61	1.81	0.00
0.6*DL+WL1	-0.03	0.15	0.60	0.0	0.00	15.56	-0.03	1.20	0.60	0.06
0.6*DL+WL2	0.03	0.16	0.60	0.0	0.00	15.57	0.03	1.20	0.60	0.06
0.6*DL+WL3	0.00	0.25	0.34	0.0	0.00	15.66	0.00	0.68	0.34	0.00
0.6*DL+WL4	0.00	0.06	0.86	0.0	0.00	15.47	0.00	1.72	0.86	0.00
0.6*DL+0.7*EL1	-0.10	0.15	0.60	0.0	0.00	15.56	-0.10	1.20	0.60	0.20
0.6*DL+0.7*EL2	0.10	0.17	0.60	0.0	0.00	15.58	0.10	1.20	0.60	0.20
0.6*DL+0.7*EL3	0.10	0.17	0.60	0.0	0.00	15.58	0.10	1.20	0.60	0.20
0.6*DL+0.7*EL4	0.00	0.11	0.67	0.0	0.00	15.52	0.00	1.35	0.67	0.00

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	18.27	0.00	0.00	2.00	4.00	-	3.77	-	3.65
DL+0.75*L+0.75*WL1	17.62	-0.02	0.05	1.75	3.50	312.57	4.15	303.10	4.03
DL+0.75*L+0.75*WL2	17.62	0.02	0.05	1.75	3.50	312.66	4.15	303.18	4.03
DL+0.75*L+0.75*WL3	17.69	0.00	0.00	1.55	3.11	10811.27	4.69	10483.66	4.55
DL+0.75*L+0.75*WL4	17.55	0.00	0.00	1.95	3.89	10724.28	3.72	10399.30	3.61
DL+0.75*L+0.75*0.7*EL1	17.61	-0.07	0.15	1.75	3.50	98.77	4.15	95.78	4.03
DL+0.75*L+0.75*0.7*EL2	17.63	0.07	0.15	1.75	3.50	98.86	4.15	95.87	4.03
DL+0.75*L+0.75*0.7*EL3	17.63	0.07	0.15	1.75	3.50	98.86	4.15	95.87	4.03
DL+0.75*L+0.75*0.7*EL4	17.58	0.00	0.00	1.81	3.61	6578.22	4.02	6378.88	3.89
0.6*DL+WL1	15.56	-0.03	0.06	0.60	1.20	207.09	10.70	200.81	10.38
0.6*DL+WL2	15.57	0.03	0.06	0.60	1.20	207.17	10.70	200.89	10.38
0.6*DL+WL3	15.66	0.00	0.00	0.34	0.68	7177.91	19.01	6960.40	18.44
0.6*DL+WL4	15.47	0.00	0.00	0.86	1.72	7090.92	7.42	6876.04	7.19
0.6*DL+0.7*EL1	15.56	-0.10	0.20	0.60	1.20	65.43	10.72	63.45	10.40
0.6*DL+0.7*EL2	15.58	0.10	0.20	0.60	1.20	65.52	10.68	63.53	10.36
0.6*DL+0.7*EL3	15.58	0.10	0.20	0.60	1.20	65.52	10.68	63.53	10.36
0.6*DL+0.7*EL4	15.52	0.00	0.00	0.67	1.35	4353.71	9.49	4221.78	9.20

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	0.00	0.36	1.40	0.0	0.00	9.034	2.80	0.00
1.4D+1.7L	0.00	4.78	3.10	0.0	0.00	13.454	6.20	0.00
1.2D + 1.6WL1+0.5*L	-0.05	1.61	1.70	0.0	0.00	10.277	3.40	0.10
1.2D + 1.6WL2+0.5*L	0.05	1.62	1.70	0.0	0.00	10.287	3.40	0.10
1.2D + 1.6WL3+0.5*L	0.00	1.76	1.28	0.0	0.00	10.434	2.57	0.00
1.2D + 1.6WL4+0.5*L	0.00	1.46	2.12	0.0	0.00	10.130	4.23	0.00
1.2D + 1.0EL1+0.5*L	-0.14	1.60	1.70	0.0	0.00	10.267	3.40	0.28
1.2D + 1.0EL2+0.5*L	0.14	1.63	1.70	0.0	0.00	10.297	3.40	0.28
1.2D + 1.0EL3+0.5*L	0.14	1.63	1.70	0.0	0.00	10.297	3.40	0.28
1.2D + 1.0EL4+0.5*L	0.00	1.54	1.81	0.0	0.00	10.209	3.61	0.00
0.9D+1.6WL1	-0.05	0.23	0.90	0.0	0.00	8.899	1.80	0.10
0.9D+1.6WL2	0.05	0.24	0.90	0.0	0.00	8.909	1.80	0.10
0.9D+1.6WL3	0.00	0.39	0.48	0.0	0.00	9.056	0.97	0.00
0.9D+1.6WL4	0.00	0.08	1.32	0.0	0.00	8.752	2.63	0.00
0.9D+1.0EL1	-0.14	0.22	0.90	0.0	0.00	8.889	1.80	0.28
0.9D+1.0EL2	0.14	0.25	0.90	0.0	0.00	8.919	1.80	0.28
0.9D+1.0EL3	0.14	0.25	0.90	0.0	0.00	8.919	1.80	0.28
0.9D+1.0EL4	0.00	0.16	1.01	0.0	0.00	8.831	2.01	0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

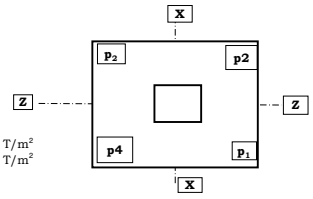
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 58.1 T/m^2 For WL/SL case = 70.6 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m^2) V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+LL	18.27	0.00	4.00	6.71	0.00	5.34	14.05	3.37	14.05	3.37	11.35	**O.K**
DL+0.75*L+0.75*WL1	17.62	0.05	3.50	6.47	0.06	4.67	13.21	3.86	13.08	3.73	10.51	**O.K**
DL+0.75*L+0.75*WL2	17.62	0.05	3.50	6.47	0.06	4.68	13.21	3.86	13.09	3.74	10.51	**O.K**
DL+0.75*L+0.75*WL3	17.69	0.00	3.11	6.50	0.00	4.15	12.65	4.35	12.65	4.34	9.95	**O.K**
DL+0.75*L+0.75*WL4	17.55	0.00	3.89	6.45	0.00	5.20	13.64	3.25	13.64	3.25	10.94	**O.K**
DL+0.75*L+0.75*0.7*EL1	17.61	0.15	3.50	6.47	0.20	4.67	13.34	3.99	12.94	3.60	10.64	**O.K**
DL+0.75*L+0.75*0.7*EL2	17.63	0.15	3.50	6.47	0.20	4.68	13.35	3.99	12.96	3.60	10.65	**O.K**
DL+0.75*L+0.75*0.7*EL3	17.63	0.15	3.50	6.47	0.20	4.68	13.35	3.99	12.96	3.60	10.65	**O.K**
DL+0.75*L+0.75*0.7*EL4	17.58	0.00	3.61	6.46	0.00	4.82	13.29	3.64	13.28	3.63	10.59	**O.K**
0.6*DL+WL1	15.56	0.06	1.20	5.72	0.08	1.60	9.40	6.20	9.24	6.03	6.70	**O.K**
0.6*DL+WL2	15.57	0.06	1.20	5.72	0.08	1.60	9.40	6.20	9.24	6.03	6.70	**O.K**
0.6*DL+WL3	15.66	0.00	0.68	5.75	0.00	0.91	8.66	6.85	8.66	6.84	5.96	**O.K**
0.6*DL+WL4	15.47	0.00	1.72	5.68	0.00	2.30	9.98	5.39	9.98	5.38	7.28	**O.K**
0.6*DL+0.7*EL1	15.56	0.20	1.20	5.71	0.26	1.60	9.57	6.38	9.05	5.85	6.87	**O.K**
0.6*DL+0.7*EL2	15.58	0.20	1.20	5.72	0.26	1.61	9.59	6.38	9.07	5.85	6.89	**O.K**
0.6*DL+0.7*EL3	15.58	0.20	1.20	5.72	0.26	1.61	9.59	6.38	9.07	5.85	6.89	**O.K**
0.6*DL+0.7*EL4	15.52	0.00	1.35	5.70	0.00	1.80	9.51	5.90	9.50	5.89	6.81	**O.K**

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

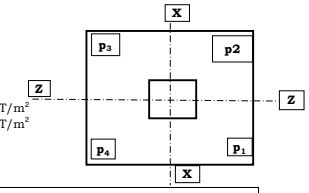
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 58.1 T/m^2 For WL/SL case = 70.6 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4D	9.0	0.00	2.80	3.32	0.00	3.74	9.06	1.58	9.06	1.58	6.36
1.4D+1.7L	13.5	0.00	6.20	4.94	0.00	8.28	15.22	-1.34	15.22	-1.34	12.52
1.2D + 1.6WL1+0.5*L	10.3	0.10	3.40	3.77	0.13	4.54	10.45	1.37	10.18	1.10	7.75
1.2D + 1.6WL2+0.5*L	10.3	0.10	3.40	3.78	0.13	4.54	10.45	1.37	10.19	1.10	7.75
1.2D + 1.6WL3+0.5*L	10.4	0.00	2.57	3.83	0.00	3.43	9.26	2.41	9.26	2.40	6.56
1.2D + 1.6WL4+0.5*L	10.1	0.00	4.23	3.72	0.00	5.65	11.38	0.07	11.37	0.06	8.68
1.2D + 1.0EL1+0.5*L	10.3	0.28	3.40	3.77	0.37	4.53	10.68	1.61	9.93	0.86	7.98
1.2D + 1.0EL2+0.5*L	10.3	0.28	3.40	3.78	0.37	4.55	10.70	1.61	9.96	0.86	8.00
1.2D + 1.0EL3+0.5*L	10.3	0.28	3.40	3.78	0.37	4.55	10.70	1.61	9.96	0.86	8.00
1.2D + 1.0EL4+0.5*L	10.2	0.00	3.61	3.75	0.01	4.83	10.58	0.93	10.57	0.92	7.88
0.9D+1.6WL1	8.9	0.10	1.80	3.27	0.13	2.40	7.80	3.00	7.54	2.73	5.10
0.9D+1.6WL2	8.9	0.10	1.80	3.27	0.13	2.40	7.81	3.00	7.54	2.74	5.11
0.9D+1.6WL3	9.1	0.00	0.97	3.33	0.00	1.29	6.62	4.04	6.61	4.03	3.92
0.9D+1.6WL4	8.8	0.00	2.63	3.21	0.00	3.52	8.73	1.70	8.73	1.69	6.03
0.9D+1.0EL1	8.9	0.28	1.80	3.27	0.37	2.40	8.04	3.24	7.29	2.49	5.34
0.9D+1.0EL2	8.9	0.28	1.80	3.28	0.37	2.41	8.06	3.24	7.31	2.49	5.36
0.9D+1.0EL3	8.9	0.28	1.80	3.28	0.37	2.41	8.06	3.24	7.31	2.49	5.36
0.9D+1.0EL4	8.8	0.00	2.01	3.24	0.01	2.69	7.94	2.56	7.93	2.55	5.24

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.65+2*0.5*0.305)+(0.65+2*0.5*0.305))*2

= 3.22 m

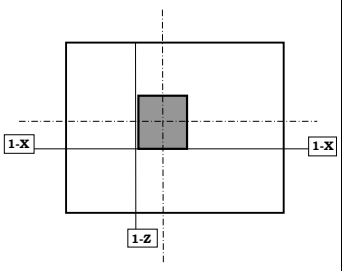
Punching Area= ((0.6+2*0.5*0.205)*(0.6+2*0.5*0.205))

= 0.65 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 2.72 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 38.69 T

Strength of Concrete in shear Vc1 = 141.18 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 147.96 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 130.17 T **Table 22.6.5.2_ (c)**

Design Vc= 130.17 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (38.69*10000)/(3220*205)

= 0.59 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear ($0.75*\sqrt{(fc)}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	6.36	1.45	2.26	1.45	2.35	19.65	0.17	OK	0.74	0.077	OK	0.16
1.4D+1.7L	12.52	2.85	4.46	2.85	4.63	38.69	0.33	OK	0.74	0.152	OK	0.31
1.2D + 1.6WL1+0.5*L	7.75	1.77	2.76	1.77	2.87	23.95	0.21	OK	0.74	0.094	OK	0.19
1.2D + 1.6WL2+0.5*L	7.75	1.77	2.76	1.77	2.87	23.95	0.21	OK	0.74	0.094	OK	0.19
1.2D + 1.6WL3+0.5*L	6.56	1.49	2.34	1.49	2.43	20.27	0.17	OK	0.74	0.08	OK	0.16
1.2D + 1.6WL4+0.5*L	8.68	1.98	3.09	1.98	3.21	26.82	0.23	OK	0.74	0.105	OK	0.21
1.2D + 1.0EL1+0.5*L	7.98	1.82	2.84	1.82	2.95	24.66	0.21	OK	0.74	0.097	OK	0.20
1.2D + 1.0EL2+0.5*L	8.00	1.82	2.85	1.82	2.96	24.72	0.21	OK	0.74	0.097	OK	0.20
1.2D + 1.0EL3+0.5*L	8.00	1.82	2.85	1.82	2.96	24.72	0.21	OK	0.74	0.097	OK	0.20
1.2D + 1.0EL4+0.5*L	7.88	1.80	2.81	1.80	2.92	24.35	0.21	OK	0.74	0.096	OK	0.19
0.9D+1.6WL1	5.10	1.16	1.82	1.16	1.89	15.76	0.14	OK	0.74	0.062	OK	0.12
0.9D+1.6WL2	5.11	1.16	1.82	1.16	1.89	15.79	0.14	OK	0.74	0.062	OK	0.13
0.9D+1.6WL3	3.92	0.89	1.40	0.89	1.45	12.11	0.1	OK	0.74	0.048	OK	0.10
0.9D+1.6WL4	6.03	1.37	2.15	1.37	2.23	18.63	0.16	OK	0.74	0.073	OK	0.15
0.9D+1.0EL1	5.34	1.22	1.90	1.22	1.98	16.50	0.14	OK	0.74	0.065	OK	0.13
0.9D+1.0EL2	5.36	1.22	1.91	1.22	1.98	16.56	0.14	OK	0.74	0.065	OK	0.13
0.9D+1.0EL3	5.36	1.22	1.91	1.22	1.98	16.56	0.14	OK	0.74	0.065	OK	0.13
0.9D+1.0EL4	5.24	1.19	1.87	1.19	1.94	16.19	0.14	OK	0.74	0.064	OK	0.13

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.21	0.001	219
1.4D+1.7L	0.41	0.001	219
1.2D + 1.6WL1+0.5*L	0.25	0.001	219
1.2D + 1.6WL2+0.5*L	0.25	0.001	219
1.2D + 1.6WL3+0.5*L	0.21	0.001	219
1.2D + 1.6WL4+0.5*L	0.28	0.001	219
1.2D + 1.0EL1+0.5*L	0.26	0.001	219
1.2D + 1.0EL2+0.5*L	0.26	0.001	219
1.2D + 1.0EL3+0.5*L	0.26	0.001	219
1.2D + 1.0EL4+0.5*L	0.26	0.001	219
0.9D+1.6WL1	0.17	0	0
0.9D+1.6WL2	0.17	0	0
0.9D+1.6WL3	0.13	0	0
0.9D+1.6WL4	0.2	0	0
0.9D+1.0EL1	0.17	0	0
0.9D+1.0EL2	0.18	0	0
0.9D+1.0EL3	0.18	0	0
0.9D+1.0EL4	0.17	0	0

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd. (mm)
At bottom	2.85	0.31	0.41	0.001	219.00	394.20	12ϕT	286.90	300	200
At top	0.00	0.00	0.00	0.0018	369.00	369.00	12ϕT	306.50	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

4.4 Design of Foundation mkd. F4

	1			2		
LOAD ITEM	Fx (T)	Fy (T)	Fz (T)	Fx (T)	Fy (T)	Fz (T)
1 DL	0.00	0.50	-0.06	0.00	0.50	-0.06
2 LL	0.00	3.00	-0.06	0.00	3.00	-0.06
3 WL1(+X)	0.00	-0.01	0.00	0.00	0.01	0.00
4 WL2(-X)	0.00	0.01	0.00	0.00	-0.01	0.00
5 WL3(+Z)	0.00	0.06	-0.11	0.00	0.07	-0.10
6 WL4(-Z)	0.00	-0.06	0.11	0.00	-0.07	0.10
7 SL1(+X)	-0.02	-0.05	0.00	-0.02	0.06	-0.01
8 SL2(-X)	0.02	0.05	0.00	0.02	-0.06	0.01
9 SL3(+Z)	0.02	0.05	0.00	0.02	-0.06	0.01
10 SL4(-Z)	0.00	-0.06	0.02	0.00	-0.06	0.02

FOUNDATION SIZE 1.00m X 3.75m
DEPTH OF THE FOUNDATION 1.10m (below FGL)
THE WT OF SOIL +SELF WT 1X3.75X1.1XUNIT WT = 8.25 T

Calculation of Platform load

Dead Load Intensity = 0.15 T/m^2 (grating wt = 0.05T/m2)
Live Load Intensity = 1 T/m^2
Width of Platform = 2.75 m
Influence length of Platform = (1.225+1.88*0.5) = 2.215m
Influence area of platform = 2.215*2.75/2 m^2
3.0 m^2
Dead load on column = 0.457 T
Live load on column = 3.0 T

For loads other than Dead & Live Load refer Load Data For Foundation mkd. F2 & F2A

WORKING STRESS DESIGN

	C-1			C-2			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	3.50	-0.11	0.00	3.50	-0.11	15.25	0.00	0.36	-0.23	0.00
DL+0.75*L+0.75*WL1	0.00	10.89	-0.10	-0.01	10.89	-0.10	30.03	0.00	0.32	-0.20	0.02
DL+0.75*L+0.75*WL2	0.01	10.91	-0.10	0.00	10.87	-0.10	30.03	0.01	0.32	-0.20	0.06
DL+0.75*L+0.75*WL3	0.00	10.94	-0.18	-0.01	10.93	-0.18	30.13	-0.01	0.57	-0.36	0.02
DL+0.75*L+0.75*WL4	0.01	10.86	-0.02	0.00	10.82	-0.02	29.93	0.01	0.06	-0.04	0.05
DL+0.75*L+0.75*0.7*EL1	-0.01	10.87	-0.10	-0.01	10.91	-0.10	30.03	-0.02	0.32	-0.20	0.08
DL+0.75*L+0.75*0.7*EL2	0.01	10.93	-0.10	0.01	10.85	-0.10	30.03	0.02	0.31	-0.20	0.14
DL+0.75*L+0.75*0.7*EL3	0.01	10.93	-0.10	0.01	10.85	-0.10	30.03	0.02	0.31	-0.20	0.14
DL+0.75*L+0.75*0.7*EL4	0.00	10.87	-0.09	0.00	10.85	-0.09	29.97	0.00	0.28	-0.17	0.03
0.6*DL+WL1	0.00	0.91	-0.03	-0.01	0.91	-0.03	10.07	-0.01	0.11	-0.07	0.02
0.6*DL+WL2	0.01	0.93	-0.03	0.00	0.89	-0.03	10.07	0.01	0.10	-0.07	0.06
0.6*DL+WL3	0.00	0.97	-0.14	-0.01	0.97	-0.13	10.20	-0.01	0.44	-0.28	0.02
0.6*DL+WL4	0.01	0.86	0.08	0.00	0.83	0.07	9.94	0.01	0.23	0.15	0.05
0.6*DL+0.7*EL1	-0.01	0.9	0.0	-0.02	0.94	-0.04	10.07	-0.03	0.11	-0.07	0.13
0.6*DL+0.7*EL2	0.02	1.0	0.0	0.01	0.86	-0.03	10.06	0.03	0.10	-0.06	0.17
0.6*DL+0.7*EL3	0.02	1.0	0.0	0.01	0.86	-0.03	10.06	0.03	0.10	-0.06	0.17
0.6*DL+0.7*EL4	0.00	0.9	0.0	0.00	0.86	-0.02	9.98	0.00	0.05	-0.03	0.02

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	15.25	0.00	0.00	-0.23	0.36	22338.87	20.98	7625.00	26.86
DL+0.75*L+0.75*WL1	30.03	0.00	0.02	-0.20	0.32	3660.59	47.29	2669.25	60.53
DL+0.75*L+0.75*WL2	30.03	0.01	0.06	-0.20	0.32	953.17	47.54	2287.86	60.85
DL+0.75*L+0.75*WL3	30.13	-0.01	0.02	-0.36	0.57	2302.11	26.39	2086.66	33.78
DL+0.75*L+0.75*WL4	29.93	0.01	0.05	-0.04	0.06	1039.40	-	1834.85	305.81
DL+0.75*L+0.75*0.7*EL1	30.03	-0.02	0.08	-0.20	0.32	678.03	46.78	622.00	59.88
DL+0.75*L+0.75*0.7*EL2	30.03	0.02	0.14	-0.20	0.31	393.43	48.06	598.65	61.51
DL+0.75*L+0.75*0.7*EL3	30.03	0.02	0.14	-0.20	0.31	393.43	48.06	-	61.51
DL+0.75*L+0.75*0.7*EL4	29.97	0.00	0.03	-0.17	0.28	1638.36	54.32	-	69.52
0.6*DL+WL1	10.07	-0.01	0.02	-0.07	0.11	860.63	47.00	-	60.16
0.6*DL+WL2	10.07	0.01	0.06	-0.07	0.10	339.99	48.00	-	61.44
0.6*DL+WL3	10.20	-0.01	0.02	-0.28	0.44	1260.66	11.47	-	14.68
0.6*DL+WL4	9.94	0.01	0.05	0.15	0.23	382.21	21.35	-	27.33
0.6*DL+0.7*EL1	10.07	-0.03	0.13	-0.07	0.11	140.94	45.08	-	57.70
0.6*DL+0.7*EL2	10.06	0.03	0.17	-0.06	0.10	112.62	50.18	-	64.23
0.6*DL+0.7*EL3	10.06	0.03	0.17	-0.06	0.10	112.62	50.18	-	64.23
0.6*DL+0.7*EL4	9.98	0.00	0.02	-0.03	0.05	826.72	-	-	123.95

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 8 T/m^2 IN LC DL+LL . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 117) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

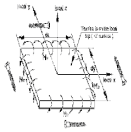
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.465	T-M	Corresponding Mxy	-0.05	T-M	*(Plate No. - 159 , L/C No. - 101)*
Min. Mx	-0.006	T-M	Corresponding Mxy	0.00	T-M	*(Plate No. - 97 , L/C No. - 101)*
Max. My	0.38	T-M	Corresponding Mxy	0.07	T-M	*(Plate No. - 166 , L/C No. - 101)*
Min. My	-1.868	T-M	Corresponding Mxy	0.00	T-M	*(Plate No. - 97 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	0.465 + 0.052 =	0.517 T-M
My1 =	My + abs(Mxy) =	0.38 + 0.066 =	0.446 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	0.465 + 0.007 =	0.472 T-M
My2 =	My+abs(Mxy^2/Mx) =	0.38 + 0.009 =	0.389 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 0.517 T-m
Myd = 0.446 T-m

ALONG X - DIRECTION :

Mu = 0.517 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 0.517X10^7/(1000X207^2) = 0.121 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast = 0.0003X1000X207/100 = 0.62 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 0.446 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 0.446X10^7/(1000X219^2) = 0.093 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X219/100 = 0.44 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-0.006 - 0.003 =	-0.009 T-M
My1 =	My - abs(Mxy) =	-1.868 - 0.003 =	-1.871 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-0.006 - 0 =	-0.006 T-M
My2 =	My-abs(Mxy^2/Mx) =	-1.868 - 0.002 =	-1.867 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -0.009 T-M
Myd = -1.871 T-M

ALONG X - DIRECTION :

Mu = 0.009 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 0.009X10^7/(1000X207^2) = 0.002 N/mm^2 < 6.9 N/mm^2
Pt = 0.0000 Reqd. Ast = 0X1000X207/100 = 0 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.871 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.871X10^7/(1000X219^2) = 0.39 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X219/100 = 2.19 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.005 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 85 , L/C No. - 101)*

Vu = 0.104 T

Provided Pt = 0.0027

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0027X0.104X10^4X207/0.009/10^7)X1000X207/10^4

= 21.867 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0027)X1000X207/10^4

= 20.544 T

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X207/10^4

= 35.514 T

HENCE, Vc = 20.544 T

Φt*Vc = 0.75X20.544 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 15.408 T **> 0.104T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.069 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 85 , L/C No. - 101)*

Vu = 1.511 T

Pt provided = 0.0026

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X1.511X10^4X219/1.871/10^7)X1000X219/10^4

= 20.901 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4

= 21.698 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X219/10^4

= 37.573 T

HENCE, Vc = 20.901 T

Φt*Vc = 0.75X20.901 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 15.676 T **> 1.511T , Hence O.K.**

PEDESTAL REINFORCEMENT

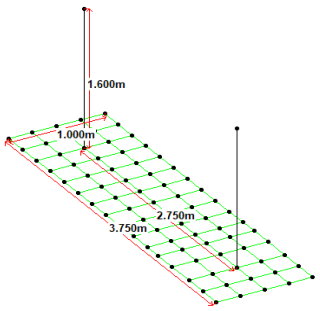
Size of Column = 500 X 325 mm^2

Reqired area of Reinforcement = 1625 mm^2

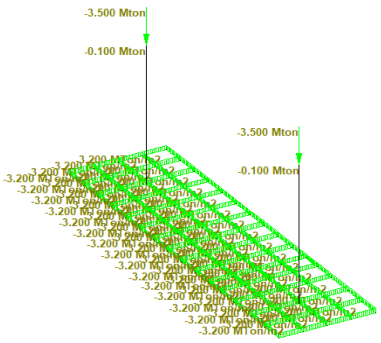
Reinforcement Provided = 4 no. 20 dia bars + 2 no. 20 dia bars

Area of Reinforcement Provided = 1885.0 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2

Soil Weight over the foundation = 1.2x 1.8 T/m2 = 2.16 T/m^2

SOIL+SURCHARGE LOAD = 3.16 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

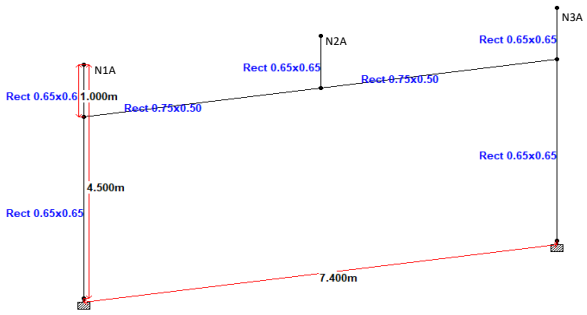
Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

4.5 Design of RC Frame mkd. FR-1 (Refer GA Dwg No. MD-216-49PR-EG-CV-GAD-2011)

Foundation below support mkd. N2A cannot be provided due to routing of Electrical cable duct. Hence a common RC Frame need to be designed to support the base of Electrical Room at Node N1A, N2A & N3A. The said Frame has been modelled & analysed in Staad-Pro. Support Reaction for the above mentioned nodes also applied to the frame. RC beam & columns are designed for the most critical Ultimate load combination in Staad. Foundations below Node N1A & N3A have been checked for the most critical support reactions from Staad Output. Foundation size & Reinforcement remained same as that of F1.

Load Detail	Col Mkd. N1A			Col Mkd. N2A			Col Mkd. N3A		
	FX (T)	FY (T)	FZ (T)	FX (T)	FY (T)	FZ (T)	FX (T)	FY (T)	FZ (T)
1 DL	0.0	2.4	0.0	0.0	4.5	0.7	0.0	4.6	1.0
2 LL	0.0	7.0	0.0	0.0	6.0	0.6	0.0	7.0	1.0
3 WL1(+X)	-0.1	-0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0
4 WL2(-X)	0.1	0.1	0.0	0.0	-0.1	0.0	0.0	0.0	0.0
5 WL3(+Z)	0.0	-0.2	-0.4	0.0	-0.1	-0.1	0.0	-0.1	-0.4
6 WL4(-Z)	0.0	0.2	0.4	0.0	0.1	0.1	0.0	0.1	0.1
7 SL1(+X)	-0.1	-0.1	0.0	-0.1	0.2	0.0	-0.1	-0.1	0.0
8 SL2(-X)	0.1	0.1	0.0	0.1	-0.2	0.0	0.1	0.1	0.0
9 SL3(+Z)	0.1	0.1	0.0	0.1	-0.2	0.0	0.1	0.1	0.0
10 SL4(-Z)	0.0	0.1	0.1	0.0	0.2	0.2	-0.1	0.1	0.2

(Refer Structural GA Dwg No. MD-216-49PR-EG-ST-GAD-V514-2008 to 2010 for Foundation Load Data)



3D MODEL OF REINFORCED CONCRETE FRAME IN STAAD-PRO

TIE BEAM (TB1) - 500 WIDE X 750 DEEP

ALL COLUMNS ARE 650 X 650 MM^2 IN SIZE

All the Reinforced Concrete members except Foundation are designed in STAAD with the following factored load combinations as per Cl. No. 2.3.2, SBC 301:2007

LOAD COMB 100 1.4D
LOAD COMB 101 1.4D+1.7L
LOAD COMB 102 1.2D + 1.6WL1+0.5*L
LOAD COMB 103 1.2D + 1.6WL2+0.5*L
LOAD COMB 104 1.2D + 1.6WL3+0.5*L
LOAD COMB 105 1.2D + 1.6WL4+0.5*L
LOAD COMB 106 1.2D + 1.0EL1+0.5*L
LOAD COMB 107 1.2D + 1.0EL2+0.5*L
LOAD COMB 108 1.2D + 1.0EL3+0.5*L
LOAD COMB 109 1.2D + 1.0EL4+0.5*L
LOAD COMB 110 0.9D+1.6WL1
LOAD COMB 111 0.9D+1.6WL2
LOAD COMB 112 0.9D+1.6WL3
LOAD COMB 113 0.9D+1.6WL4
LOAD COMB 114 0.9D+1.0EL1
LOAD COMB 115 0.9D+1.0EL2
LOAD COMB 116 0.9D+1.0EL3
LOAD COMB 117 0.9D+1.0EL4

Please refer attached STAAD Output File for the following Beam & Column numbers for design

MEMBER SCHEDULE :

Member mkd.	Member No.	Size	Reinforcement Details
TB1	175, 176	500X750	4 No. - D20 both at Top & Bottom
			Additional 3no.- D16 bars on each vertical face
			Shear Reinf. - 2L - 10 dia @ 300c/c
COLUMNS	133, 134, 173, 174	650X650	Longitudinal Rebar 4 no. D25 + 8 No. D20

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

DESIGN OF FOUNDATION MKD. F1

	ABOUT-X	ABOUT-Z
Size of column :	0.65m	0.65m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	3.00 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

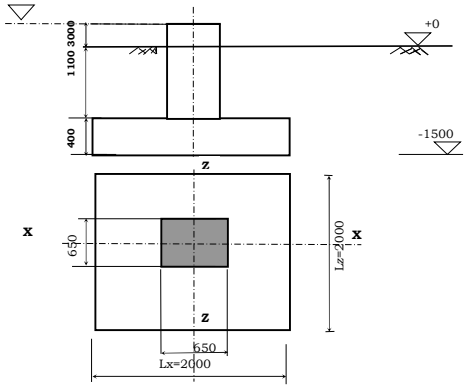
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Monorail Load	MR

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.4*2.5)+(0.65*0.65*4.1*2.5) = 8.33T

Weight of soil = ((2*2)-(0.65*0.65))*1.1*1.8 = 7.08T

U/S Base Pl Total = 15.41T



Design Input:

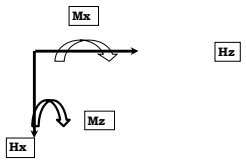
Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 319mm = 0.319m

Effective depth of slab along Z direction = 305mm = 0.305m

Clear projection along Z direction = 0.675m

Clear projection along X direction = 0.675m



WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
D+L	-4.83	25.91	-2.51	-10.6	5.26	41.32	-4.83	10.61	-2.51	5.26
DL+0.75*L+0.75*WL1	-4.18	22.88	-2.23	-9.4	4.48	38.29	-4.18	9.42	-2.23	4.48
DL+0.75*L+0.75*WL2	-4.31	22.94	-2.23	-9.4	4.75	38.35	-4.31	9.42	-2.23	4.75
DL+0.75*L+0.75*WL3	-4.24	22.91	-2.22	-9.4	4.62	38.32	-4.24	9.36	-2.22	4.62
DL+0.75*L+0.75*WL4	-4.24	22.91	-2.22	-9.4	4.62	38.32	-4.24	9.36	-2.22	4.62
DL+0.75*L+0.75*0.7*EL1	-2.36	12.58	-0.86	-3.7	2.50	27.99	-2.36	3.66	-0.86	2.50
DL+0.75*L+0.75*0.7*EL2	-2.49	12.64	-0.86	-3.7	2.77	28.05	-2.49	3.66	-0.86	2.77
DL+0.75*L+0.75*0.7*EL3	-2.42	12.61	-0.85	-3.6	2.64	28.02	-2.42	3.59	-0.85	2.64
DL+0.75*L+0.75*0.7*EL4	-2.42	12.61	-0.87	-3.7	2.64	28.02	-2.42	3.72	-0.87	2.64
0.6*DL+WL1	-4.21	22.89	-2.23	-9.4	4.55	38.30	-4.21	9.42	-2.23	4.55
0.6*DL+WL2	-4.27	22.92	-2.23	-9.4	4.69	38.33	-4.27	9.42	-2.23	4.69
0.6*DL+WL3	-4.24	22.91	-2.23	-9.4	4.62	38.32	-4.24	9.39	-2.23	4.62
0.6*DL+WL4	-4.24	22.91	-2.23	-9.5	4.62	38.32	-4.24	9.46	-2.23	4.62
0.6*DL+0.7*EL1	-2.36	12.58	-0.86	-3.7	2.50	27.99	-2.36	3.66	-0.86	2.50
0.6*DL+0.7*EL2	-2.49	12.64	-0.86	-3.7	2.77	28.05	-2.49	3.66	-0.86	2.77
0.6*DL+0.7*EL3	-2.42	12.61	-0.85	-3.6	2.64	28.02	-2.42	3.59	-0.85	2.64
0.6*DL+0.7*EL4	-2.42	12.61	-0.87	-3.7	2.64	28.02	-2.42	3.72	-0.87	2.64

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
D+L	41.32	-4.83	5.26	-2.51	10.61	7.86	3.89	3.42	6.58
DL+0.75*L+0.75*WL1	38.29	-4.18	4.48	-2.23	9.42	8.54	4.06	3.67	6.87
DL+0.75*L+0.75*WL2	38.35	-4.31	4.75	-2.23	9.42	8.07	4.07	3.56	6.88
DL+0.75*L+0.75*WL3	38.32	-4.24	4.62	-2.22	9.36	8.30	4.10	3.61	6.89
DL+0.75*L+0.75*WL4	38.32	-4.24	4.62	-2.22	9.36	8.30	4.10	3.61	6.89
DL+0.75*L+0.75*0.7*EL1	27.99	-2.36	2.50	-0.86	3.66	11.18	7.66	4.75	13.03
DL+0.75*L+0.75*0.7*EL2	28.05	-2.49	2.77	-0.86	3.66	10.11	7.67	4.51	13.06
DL+0.75*L+0.75*0.7*EL3	28.02	-2.42	2.64	-0.85	3.59	10.62	7.81	4.63	13.15
DL+0.75*L+0.75*0.7*EL4	28.02	-2.42	2.64	-0.87	3.72	10.62	7.53	4.63	12.94
0.6*DL+WL1	38.30	-4.21	4.55	-2.23	9.42	8.41	4.06	3.64	6.87
0.6*DL+WL2	38.33	-4.27	4.69	-2.23	9.42	8.18	4.07	3.59	6.88
0.6*DL+WL3	38.32	-4.24	4.62	-2.23	9.39	8.30	4.08	3.61	6.89
0.6*DL+WL4	38.32	-4.24	4.62	-2.23	9.46	8.30	4.05	3.61	6.86
0.6*DL+0.7*EL1	27.99	-2.36	2.50	-0.86	3.66	11.18	7.66	4.75	13.03
0.6*DL+0.7*EL2	28.05	-2.49	2.77	-0.86	3.66	10.11	7.67	4.51	13.06
0.6*DL+0.7*EL3	28.02	-2.42	2.64	-0.85	3.59	10.62	7.81	4.63	13.15
0.6*DL+0.7*EL4	28.02	-2.42	2.64	-0.87	3.72	10.62	7.53	4.63	12.94

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C02	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C02	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
100	-3.82	18.24	-1.80	-7.8	4.14	33.650	7.77	4.14
101	-7.12	34.24	-3.89	-16.7	7.71	49.650	16.72	7.71
102	-4.41	21.49	-2.18	-9.4	4.61	36.904	9.40	4.61
103	-4.68	21.59	-2.18	-9.4	5.24	36.997	9.40	5.24
104	-4.55	21.54	-2.17	-9.3	4.93	36.950	9.29	4.93
105	-4.55	21.54	-2.20	-9.5	4.93	36.950	9.51	4.93
106	-2.76	13.99	-1.43	-6.2	2.83	29.404	6.19	2.83
107	-3.03	14.09	-1.43	-6.2	3.45	29.497	6.19	3.45
108	-2.90	14.04	-1.42	-6.1	3.14	29.451	6.08	3.14
109	-2.90	14.04	-1.44	-6.3	3.14	29.451	6.30	3.14
110	-4.48	21.52	-2.18	-9.4	4.77	36.927	9.40	4.77
111	-4.61	21.56	-2.18	-9.4	5.08	36.974	9.40	5.08
112	-4.55	21.54	-2.18	-9.3	4.93	36.950	9.34	4.93
113	-4.55	21.54	-2.19	-9.5	4.93	36.950	9.46	4.93
114	-2.83	14.02	-1.43	-6.2	2.98	29.428	6.19	2.98
115	-2.96	14.06	-1.43	-6.2	3.29	29.474	6.19	3.29
116	-2.90	14.04	-1.42	-6.1	3.14	29.451	6.14	3.14
117	-2.90	14.04	-1.44	-6.2	3.14	29.451	6.25	3.14

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

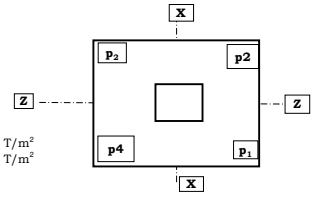
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 58.1 T/m^2 For WL/SL case = 70.6 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m^2) V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
D+L	41.32	5.26	10.61	10.33	3.94	7.96	24.23	8.32	16.34	0.43	21.53	O.K
DL+0.75*L+0.75*WL1	38.29	4.48	9.42	9.57	3.36	7.07	22.00	7.87	15.28	1.14	19.30	O.K
DL+0.75*L+0.75*WL2	38.35	4.75	9.42	9.59	3.57	7.07	22.22	8.08	15.09	0.95	19.52	O.K
DL+0.75*L+0.75*WL3	38.32	4.62	9.36	9.58	3.46	7.02	22.06	8.03	15.13	1.10	19.36	O.K
DL+0.75*L+0.75*WL4	38.32	4.62	9.36	9.58	3.46	7.02	22.06	8.03	15.13	1.10	19.36	O.K
DL+0.75*L+0.75*0.7*EL1	27.99	2.50	3.66	7.00	1.88	2.74	13.62	8.13	9.86	4.38	10.92	O.K
DL+0.75*L+0.75*0.7*EL2	28.05	2.77	3.66	7.01	2.08	2.74	13.83	8.35	9.67	4.19	11.13	O.K
DL+0.75*L+0.75*0.7*EL3	28.02	2.64	3.59	7.00	1.98	2.69	13.68	8.29	9.72	4.33	10.98	O.K
DL+0.75*L+0.75*0.7*EL4	28.02	2.64	3.72	7.00	1.98	2.79	13.77	8.19	9.82	4.23	11.07	O.K
0.6*DL+WL1	38.30	4.55	9.42	9.58	3.41	7.07	22.06	7.92	15.23	1.09	19.36	O.K
0.6*DL+WL2	38.33	4.69	9.42	9.58	3.52	7.07	22.17	8.03	15.13	1.00	19.47	O.K
0.6*DL+WL3	38.32	4.62	9.39	9.58	3.46	7.04	22.09	8.00	15.16	1.07	19.39	O.K
0.6*DL+WL4	38.32	4.62	9.46	9.58	3.46	7.09	22.14	7.95	15.21	1.02	19.44	O.K
0.6*DL+0.7*EL1	27.99	2.50	3.66	7.00	1.88	2.74	13.62	8.13	9.86	4.38	10.92	O.K
0.6*DL+0.7*EL2	28.05	2.77	3.66	7.01	2.08	2.74	13.83	8.35	9.67	4.19	11.13	O.K
0.6*DL+0.7*EL3	28.02	2.64	3.59	7.00	1.98	2.69	13.68	8.29	9.72	4.33	10.98	O.K
0.6*DL+0.7*EL4	28.02	2.64	3.72	7.00	1.98	2.79	13.77	8.19	9.82	4.23	11.07	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

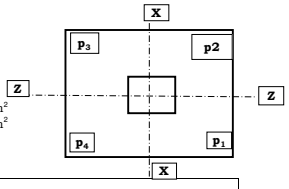
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 58.1 T/m^2 For WL/SL case = 70.6 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
100	33.7	4.14	7.77	8.41	3.11	5.83	19.35	7.69	13.13	1.48	16.65
101	49.7	7.71	16.72	12.41	5.78	12.54	32.73	5.66	19.17	-5.91	30.75
102	36.9	4.61	9.40	9.23	3.46	7.05	21.74	5.64	12.82	-1.28	19.76
103	37.0	5.24	9.40	9.25	3.93	7.05	22.23	6.13	12.37	-1.73	20.25
104	37.0	4.93	9.29	9.24	3.69	6.97	21.90	5.97	12.51	-1.42	19.92
105	37.0	4.93	9.51	9.24	3.69	7.13	22.07	5.80	12.68	-1.59	20.09
106	29.4	2.83	6.19	7.35	2.12	4.64	16.11	4.83	9.87	0.59	14.13
107	29.5	3.45	6.19	7.37	2.59	4.64	16.60	5.32	9.43	0.14	14.62
108	29.5	3.14	6.08	7.36	2.35	4.56	16.27	5.16	9.57	0.45	14.29
109	29.5	3.14	6.30	7.36	2.35	4.73	16.44	4.99	9.74	0.28	14.46
110	36.9	4.77	9.40	9.23	3.58	7.05	21.86	5.76	12.71	-1.40	19.88
111	37.0	5.08	9.40	9.24	3.81	7.05	22.10	6.00	12.48	-1.62	20.12
112	37.0	4.93	9.34	9.24	3.69	7.01	21.94	5.92	12.55	-1.46	19.96
113	37.0	4.93	9.46	9.24	3.69	7.09	22.02	5.84	12.64	-1.55	20.04
114	29.4	2.98	6.19	7.36	2.24	4.64	16.24	4.95	9.76	0.48	14.26
115	29.5	3.29	6.19	7.37	2.47	4.64	16.48	5.20	9.54	0.26	14.50
116	29.5	3.14	6.14	7.36	2.35	4.60	16.32	5.12	9.61	0.41	14.34
117	29.5	3.14	6.25	7.36	2.35	4.69	16.40	5.03	9.69	0.32	14.42

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTIO N & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

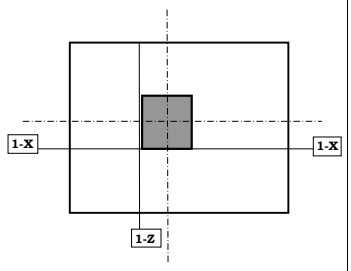
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.65+2*0.5*0.305)+(0.65+2*0.5*0.305))*2
= 3.82 m

Punching Area= ((0.65+2*0.5*0.305)*(0.65+2*0.5*0.305))
= 0.91 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*2.5+1.1*1.8)
= 2.98 T/m^2

Area of footing, A_f= 4.00 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 95.02 T
Strength of Concrete in shear Vc1 = 249.18 T **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 298.33 T **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 229.76 T **Table 22.6.5.2_ (c)**
Design Vc= 229.76 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (95.02*10000)/(3820*305)
= 0.82 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear ($0.75*\sqrt{fc}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
100	16.65	3.79	5.93	3.79	6.16	51.45	0.44	OK	0.74	0.202	OK	0.41
101	30.75	7.01	10.95	7.01	11.38	95.02	0.82	OK	0.74	0.373	OK	0.75
102	19.76	4.50	7.03	4.50	7.31	61.06	0.52	OK	0.74	0.24	OK	0.48
103	20.25	4.61	7.21	4.61	7.49	62.57	0.54	OK	0.74	0.246	OK	0.50
104	19.92	4.54	7.09	4.54	7.37	61.55	0.53	OK	0.74	0.242	OK	0.49
105	20.09	4.58	7.15	4.58	7.43	62.08	0.53	OK	0.74	0.244	OK	0.49
106	14.13	3.22	5.03	3.22	5.23	43.66	0.37	OK	0.74	0.171	OK	0.35
107	14.62	3.33	5.20	3.33	5.41	45.18	0.39	OK	0.74	0.177	OK	0.36
108	14.29	3.26	5.09	3.26	5.29	44.16	0.38	OK	0.74	0.173	OK	0.35
109	14.46	3.29	5.15	3.29	5.35	44.68	0.38	OK	0.74	0.175	OK	0.35
110	19.88	4.53	7.08	4.53	7.36	61.43	0.53	OK	0.74	0.241	OK	0.49
111	20.12	4.58	7.16	4.58	7.44	62.17	0.53	OK	0.74	0.244	OK	0.49
112	19.96	4.55	7.11	4.55	7.39	61.68	0.53	OK	0.74	0.242	OK	0.49
113	20.04	4.57	7.13	4.57	7.41	61.92	0.53	OK	0.74	0.243	OK	0.49
114	14.26	3.25	5.08	3.25	5.28	44.06	0.38	OK	0.74	0.173	OK	0.35
115	14.50	3.30	5.16	3.30	5.37	44.81	0.38	OK	0.74	0.176	OK	0.36
116	14.34	3.27	5.11	3.27	5.31	44.31	0.38	OK	0.74	0.174	OK	0.35
117	14.42	3.29	5.13	3.29	5.34	44.56	0.38	OK	0.74	0.175	OK	0.35

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
100	0.54	0.001	319
101	1	0.002	638
102	0.65	0.002	638
103	0.66	0.002	638
104	0.65	0.002	638
105	0.66	0.002	638
106	0.46	0.001	319
107	0.48	0.001	319
108	0.47	0.001	319
109	0.47	0.001	319
110	0.65	0.002	638
111	0.66	0.002	638
112	0.65	0.002	638
113	0.65	0.002	638
114	0.47	0.001	319
115	0.47	0.001	319
116	0.47	0.001	319
117	0.47	0.001	319

Location	Mux (T-m)	ϕ Kn = M_{ul}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd. (mm)
At bottom	7.01	0.75	1.00	0.002	638.00	574.20	12ϕT	177.27	300	150
At top	0.00	0.00	0.00	0.0018	549.00	549.00	12ϕT	206.01	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**150**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**150**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023			Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTIO N & REVISED AS MARKED	AS BUILT	
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI	

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] SUBSTATION STRUCTURAL DOC NO. MD-216-49PR-EG-ST-CAL-V514-2001

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3	
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023			Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTIO N & REVISED AS MARKED	AS BUILT	
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI	

ATTACHMENT 1 FOUNDATION LOAD DATA FOR SUBSTATION COLUMNS

DRAWING NUMBER: MD-216-49PR-EG-ST-GAD-V514-2008 | REV: A01

		DEAD LOAD				LIVE LOAD		
	Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN
N1A	12	0.171	18.186	-0.045	12	0.081	8.413	-0.305
N2A	23	0.089	43.294	6.942	23	0.083	57.577	5.596
N3A	27	0.017	45.883	8.919	27	0.041	69.185	7.282
N4A	35	-0.017	46.309	9.101	35	-0.024	68.919	7.321
N5A	31	-0.008	42.745	6.995	31	0	66.698	6.115
N6A	10	-0.283	12.97	0.154	10	-0.144	15.306	0.27
N7A	13	0.176	37.736	-0.141	13	0.128	20.665	-0.281
N8A	24	-0.069	13.277	-0.103	24	-0.02	28.401	-0.093
N9A	28	0.017	11.365	0	28	0.023	32.126	0.001
N10A	36	0.007	11.932	0.004	36	0.009	32.545	0.003
N11A	32	0.027	12.185	-0.011	32	0.031	33.07	-0.013
N12A	150	-0.224	18.298	-0.588	150	-0.301	20.978	-0.961
N1B	13	0.176	37.736	-0.141	13	0.128	20.665	-0.281
N2B	25	0.033	12.288	-0.013	25	0.035	6.574	-0.014
N3B	29	-0.015	11.155	-0.002	29	-0.014	31.701	-0.003
N4B	37	0.001	9.246	0.002	37	0.002	15.736	0.001
N5B	33	0	12.682	-0.019	33	-0.003	17.887	-0.021
N6B	151	-0.152	17.053	-0.169	151	-0.142	12.416	-0.251
N7B	15	0.397	14.611	-0.835	15	0.257	7.195	-0.646
N8B	26	-0.017	42.258	-6.275	26	0.003	26.366	-4.747
N9B	30	0.08	45.488	-8.023	30	0.163	38.604	-6.459
N10B	38	-0.067	46.124	-7.858	38	-0.09	30.04	-5.987
N11B	34	0.04	40.622	-6.063	34	0.039	27.079	-4.802
N12B	152	-0.449	13.424	-0.768	152	-0.322	7.863	-0.732
N1C	17	0.45	13.597	0.472	17	0.256	7.465	0.349
N2C	188	0.817	32.048	-0.031	188	0.707	27.642	-0.082
N3C	20	0.407	13.584	-0.442	20	0.23	6.831	-0.269
N4C	18	-0.45	14.431	0.51	18	-0.262	7.622	0.356
N5C	189	-0.817	34.722	-0.035	189	-0.697	23.458	-0.071
N6C	19	-0.407	14.126	-0.475	19	-0.235	6.826	-0.284
N1D	68	0.064	3.757	-0.034	68	0.047	1.207	-0.051
N2D	69	0.189	4.44	-0.064	69	0.101	1.818	-0.056
N1E	72	0.029	5.25	-0.551	72	0.019	2.793	-0.573
N2E	73	-0.038	5.036	-0.553	73	-0.002	2.822	-0.594

	SEISMIC+X				SEISMIC+Z				SEISMIC-X				SEISMIC-Z			
Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN
12	-1.534	-1.432	-0.071		12	1.534	1.432	0.071	12	1.534	1.432	0.071	12	0.023	0.898	0.962
23	-1.125	2.185	0.202		23	1.125	-2.185	-0.202	23	1.125	-2.185	-0.202	23	-0.103	1.631	1.825
27	-0.908	-0.497	0.092		27	0.908	0.497	-0.092	27	0.908	0.497	-0.092	27	-0.023	1.359	1.814
35	-0.83	0.476	-0.086		35	0.83	-0.476	0.086	35	0.83	-0.476	0.086	35	0.001	1.339	1.804
31	-1.005	-2.08	-0.021		31	1.005	2.08	0.021	31	1.005	2.08	0.021	31	0	1.606	1.715
10	-1.58	2.503	0.109		10	1.58	-2.503	-0.109	10	1.58	-2.503	-0.109	10	-0.017	0.853	0.557
13	-1.276	-2.975	0.031		13	1.276	2.975	-0.031	13	1.276	2.975	-0.031	13	0.006	-0.16	0.532
24	-0.905	2.058	0.136		24	0.905	-2.058	-0.136	24	0.905	-2.058	-0.136	24	0.142	-0.674	0.069
28	-0.181	-0.615	-0.002		28	0.181	0.615	0.002	28	0.181	0.615	0.002	28	-0.001	-0.915	0.002
36	-0.086	0.603	0.019		36	0.086	-0.603	-0.019	36	0.086	-0.603	-0.019	36	-0.002	-0.883	-0.001
32	-0.064	-2.054	-0.013		32	0.064	2.054	0.013	32	0.064	2.054	0.013	32	-0.009	-0.729	0.004
150	-1.276	1.719	-0.146		150	1.276	-1.719	0.146	150	1.276	-1.719	0.146	150	0.015	-0.374	0.512
13	-1.276	-2.975	0.031		13	1.276	2.975	-0.031	13	1.276	2.975	-0.031	13	0.006	-0.16	0.532
25	-0.055	2.091	-0.021		25	0.055	-2.091	0.021	25	0.055	-2.091	0.021	25	-0.006	0.343	0.003
29	-0.012	-0.601	0.028		29	0.012	0.601	-0.028	29	0.012	0.601	-0.028	29	0.004	0.501	0
37	0	0.601	-0.006		37	0	-0.601	0.006	37	0	-0.601	0.006	37	-0.001	0.461	-0.001
33	-0.064	-2.08	0.032		33	0.064	2.08	-0.032	33	0.064	2.08	-0.032	33	-0.001	0.326	0.005
151	-1.312	1.69	0.012		151	1.312	-1.69	-0.012	151	1.312	-1.69	-0.012	151	-0.008	0.257	0.5
15	-1.036	-2.246	0.212		15	1.036	2.246	-0.212	15	1.036	2.246	-0.212	15	-0.011	-0.97	0.59
26	-1.585	0.648	-0.001		26	1.585	-0.648	0.001	26	1.585	-0.648	0.001	26	-0.011	-0.988	0.949
30	-1.401	-0.15	-0.024		30	1.401	0.15	0.024	30	1.401	0.15	0.024	30	0.021	-0.728	1.067
38	-1.422	0.109	0.029		38	1.422	-0.109	-0.029	38	1.422	-0.109	-0.029	38	-0.005	-0.152	1.064
34	-1.529	-0.68	0.005		34	1.529	0.68	-0.005	34	1.529	0.68	-0.005	34	-0.004	-0.86	0.935
152	-1.059	2.122	-0.138		152	1.059	-2.122	0.138	152	1.059	-2.122	0.138	152	0.014	-0.946	0.567
17	-0.621	-1.31	-0.106		17	0.621	1.31	0.106	17	0.621	1.31	0.106	17	-0.002	0.719	0.539
188	-0.894	-1.932	0.033		188	0.894	1.932	-0.033	188	0.894	1.932	-0.033	188	0.001	-0.002	0.567
20	-0.625	-1.377	0.103		20	0.625	1.377	-0.103	20	0.625	1.377	-0.103	20	0.001	-0.718	0.539
18	-0.629	1.31	0.108		18	0.629	-1.31	-0.108	18	0.629	-1.31	-0.108	18	0.002	0.72	0.539
189	-0.936	1.932	-0.044		189	0.936	-1.932	0.044	189	0.936	-1.932	0.044	189	-0.001	-0.002	0.567
19	-0.68	1.377	-0.094		19	0.68	-1.377	0.094	19	0.68	-1.377	0.094	19	-0.001	-0.717	0.539
68	-0.386	-1.133	-0.162		68	0.386	1.133	0.162	68	0.386	1.133	0.162	68	-0.012	0.539	0.122
69	-0.601	-0.305	-0.165		69	0.601	0.305	0.165	69	0.601	0.305	0.165	69	-0.02	-0.548	0.122
72	-0.19	-0.542	0.003		72	0.19	0.542	-0.003	72	0.19	0.542	-0.003	72	0.003	-0.568	0.245
73	-0.185	0.588	-0.054		73	0.185	-0.588	0.054	73	0.185	-0.588	0.054	73	0.004	-0.619	0.241

FOUNDATION REACTIONS 1

FOR NODE LOCATIONS REFER DRAWING NO: SASL-RS-DWG-ST-20K052-000105

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	12/10/2020	FIRST ISSUE	---	SGC	---	PRI	

SIEMENS Ingenuity for life

DRAWING NUMBER: SASL-RS-DWG-ST-20K052-000106 | REVISION: 1

DATE	28/09/2020	DATE	28/09/2020	DATE	28/09/2020
DRAWN	VG	CHECKED	FS	APPROVED	SJT

LARSEN & TOUBRO DRAWING NUMBER: 019045_008_S_G_26162_008 | REV: 0

OUTOTEC DRAWING NUMBER: 900707_GAU01_WG01_MTL01_45135 | DOCUMENT ID: OU100703347 | REV: 0

Outotec
The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

LARSEN & TOUBRO
This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such persons or body without our prior permission in writing.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWN	---	---
CHECKED	SGC	12/10/2020
ENGINEER	---	---
APPROVED	PRI	12/10/2020
EMPLOYER		
SCALE		A3

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
POWER GENERATION : POWER PLANT SUBSTATION
FOUNDATION LOAD DATA

DRAWING NUMBER: MD-216-49PR-EG-ST-GAD-V514-2008 | REV: A01

DRAWING NUMBER: MD-216-49PR-EG-ST-GAD-V514-2009 | REV: A01

	WINDWARD+X				WINDWARD-X				WINDWARD+Z				WINDWARD-Z			
Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN
12	-0.229	-0.315	-0.025		12	0	0	0	12	-0.034	-0.209	-0.204	12	0.05	0.629	0.662
23	-0.172	0.329	0.032		23	0	0	0	23	0.016	-0.687	-2.086	23	-0.083	1.315	1.395
27	-0.138	-0.066	0.02		27	0	0	0	27	-0.002	-1.128	-3.775	27	-0.025	0.72	1.05
35	-0.126	0.066	-0.015		35	0	0	0	35	0.009	-1.022	-3.712	35	0.001	0.654	0.915
31	-0.149	-0.31	-0.004		31	0	0	0	31	0.004	-0.707	-2.054	31	0.004	0.83	0.868
10	-0.237	0.38	0.014		10	0	0	0	10	0.013	-0.206	-0.13	10	0.008	0.578	0.376
13	-1.674	-0.526	-0.001		13	0	0	0	13	-0.01	0.056	-0.115	13	-0.003	-0.254	0.371
24	-0.149	0.336	0.023		24	0	0	0	24	-0.022	0.4	-0.018	24	0.094	-0.475	0.05
28	-0.028	-0.101	0		28	0	0	0	28	0.003	0.632	0.001	28	-0.002	-0.567	0.001
36	-0.012	0.098	0.002		36	0	0	0	36	0.001	0.618	0.002	36	-0.003	-0.461	-0.001
32	-0.003	-0.333	-0.003		32	0	0	0	32	0.008	0.406	-0.003	32	-0.005	-0.375	0.001
150	-0.214	0.277	-0.011		150	0	0	0	150	-0.015	0.092	-0.124	150	-0.001	-0.247	0.333
13	-1.674	-0.526	-0.001		13	0	0	0	13	-0.01	0.056	-0.115	13	-0.003	-0.254	0.371
25	-0.015	0.343	-0.002		25	0	0	0	25	0.009	-0.18	-0.004	25	0.003	0.272	0.002
29	-0.003	-0.097	0.004		29	0	0	0	29	-0.005	-0.352	0	29	0.003	0.31	-0.006
37	0	0.098	-0.001		37	0	0	0	37	0.001	-0.311	0	37	-0.002	0.248	-0.003
33	-0.013	-0.338	0.003		33	0	0	0	33	0.001	-0.17	-0.006	33	-0.008	0.158	0.003
151	-0.211	0.277	0.014		151	0	0	0	151	0.008	-0.062	-0.114	151	-0.011	0.154	0.332
15	-1.685	-0.472	0.028		15	0	0	0	15	0.012	0.223	-0.122	15	-0.02	-0.64	1.594
26	-0.344	0.149	0.001		26	0	0	0	26	0.021	0.506	-0.483	26	-0.002	-0.691	3.104
30	-0.31	-0.032	-0.001		30	0	0	0	30	-0.015	0.518	-0.765	30	-0.024	-0.431	1.837
38	-0.306	0.029	0.004		38	0	0	0	38	0.01	0.069	-0.734	38	0.01	-0.254	0.581
34	-0.326	-0.143	-0.002		34	0	0	0	34	-0.002	0.435	-0.473	34	-0.022	-0.441	1.672
152	-0.233	0.443	-0.022		152	0	0	0	152	-0.009	0.235	-0.136	152	-0.005	-0.57	1.562
17	0	0	0		17	0.844	1.772	0.104	17	0.002	-0.563	-1.638	17	-0.001	0.585	0.429
188	0	0	0		188	1.198	2.603	-0.032	188	-0.001	-0.052	-0.473	188	0.002	-0.052	0.454
20	0	0	0		20	0.832	1.804	-0.099	20	-0.001	0.614	-0.45	20	-0.001	-0.533	1.621
18	0	0	0		18	2.307	-1.772	-0.106	18	-0.002	-0.563	-1.637	18	0.002	0.585	0.429
189	0	0	0		189	4.198	-2.604	0.044	189	0.001	-0.051	-0.473	189	-0.002	-0.053	0.454
19	0	0	0		19	2.342	-1.803	0.089	19	0.001	0.614	-0.45	19	0.001	-0.533	1.62
68	-0.531	-0.12	-0.025		68	0	0	0	68	0.01	-0.091	-0.019	68	-0.013	0.294	0.064
69	-0.563	0.023	-0.026		69	0	0	0	69	-0.001	0.09	-0.02	69	-0.015	-0.3	0.065
72	-0.033	-0.1	0.005		72	0	0	0	72	-0.005	0.394	-0.167	72	0.037	-0.188	0.933
73	-0.032	0.106	-0.012		73	0	0	0	73	-0.006	0.452	-0.168	73	0.034	-0.267	0.849

FOUNDATION REACTIONS 2

FOR NODE LOCATIONS REFER DRAWING NO: SASL-RS-DWG-ST-20K052-000105

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	12/10/2020	FIRST ISSUE	---	SGC	---	PRI	

SIEMENS Ingenuity for life

DRAWING NUMBER: SASL-RS-DWG-ST-20K052-000107 | REVISION: 1

DRAWN	DATE	CHECKED	DATE	APPROVED	DATE
VG	07/09/2020	FS	07/09/2020	SJT	07/09/2020

LARSEN & TOUBRO DRAWING NUMBER: 019045_008_S_G_26162_009 | REV: 0

OUTOTEC DRAWING NUMBER: 900707_GAU01_WG01_MTL01_45136 | DOCUMENT ID: OU100703348 | REV: 0

Outotec
The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

LARSEN & TOUBRO
This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive. It not be copied or otherwise made use either in full or its part by such persons or body without our prior permission in writing.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWN	---	---	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SGC	12/10/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	---	---	POWER GENERATION : POWER PLANT SUBSTATION
APPROVED	PRI	12/10/2020	FOUNDATION LOAD DATA
EMPLOYER			
SCALE		A3	

DRAWING NUMBER: MD-216-49PR-EG-ST-GAD-V514-2009 | REV: A01

DRAWING NUMBER	REV
MD-216-49PR-EG-ST-GAD-V514-2010	A01

	LEEWARD+X				LEEWARD-X				LEEWARD+Z				LEEWARD-Z			
Node	Fx kN	Fy kN	Fz kN		Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN	Node	Fx kN	Fy kN	Fz kN
12	0	0	0		12	0.229	0.315	0.025	12	-0.05	-0.629	-0.662	12	0.034	0.209	0.204
23	0	0	0		23	0.172	-0.329	-0.032	23	0.083	-1.315	-1.395	23	-0.016	0.687	2.086
27	0	0	0		27	0.138	0.066	-0.02	27	0.025	-0.72	-1.05	27	0.002	1.128	3.775
35	0	0	0		35	0.126	-0.066	0.015	35	-0.001	-0.654	-0.915	35	-0.009	1.022	3.712
31	0	0	0		31	0.149	0.31	0.004	31	-0.004	-0.83	-0.868	31	-0.004	0.707	2.054
10	0	0	0		10	0.237	-0.38	-0.014	10	-0.008	-0.578	-0.376	10	-0.013	0.206	0.13
13	0	0	0		13	1.674	0.526	0.001	13	0.003	0.254	-0.371	13	0.01	-0.056	0.115
24	0	0	0		24	0.149	-0.336	-0.023	24	-0.094	0.475	-0.05	24	0.022	-0.4	0.018
28	0	0	0		28	0.028	0.101	0	28	0.002	0.567	-0.001	28	-0.003	-0.632	-0.001
36	0	0	0		36	0.012	-0.098	-0.002	36	0.003	0.461	0.001	36	-0.001	-0.618	-0.002
32	0	0	0		32	0.003	0.333	0.003	32	0.005	0.375	-0.001	32	-0.008	-0.406	0.003
150	0	0	0		150	0.214	-0.277	0.011	150	0.001	0.247	-0.333	150	0.015	-0.092	0.124
13	0	0	0		13	1.674	0.526	0.001	13	0.003	0.254	-0.371	13	0.01	-0.056	0.115
25	0	0	0		25	0.015	-0.343	0.002	25	-0.003	-0.272	-0.002	25	-0.009	0.18	0.004
29	0	0	0		29	0.003	0.097	-0.004	29	-0.003	-0.31	0.006	29	0.005	0.352	0
37	0	0	0		37	0	-0.098	0.001	37	0.002	-0.248	0.003	37	-0.001	0.311	0
33	0	0	0		33	0.013	0.338	-0.003	33	0.008	-0.158	-0.003	33	-0.001	0.17	0.006
151	0	0	0		151	0.211	-0.277	-0.014	151	0.011	-0.154	-0.332	151	-0.008	0.062	0.114
15	0	0	0		15	1.685	0.472	-0.028	15	0.02	0.64	-1.594	15	-0.012	-0.223	0.122
26	0	0	0		26	0.344	-0.149	-0.001	26	0.002	0.691	-3.104	26	-0.021	-0.506	0.483
30	0	0	0		30	0.31	0.032	0.001	30	0.024	0.431	-1.837	30	0.015	-0.518	0.765
38	0	0	0		38	0.306	-0.029	-0.004	38	-0.01	0.254	-0.581	38	-0.01	-0.069	0.734
34	0	0	0		34	0.326	0.143	0.002	34	0.022	0.441	-1.672	34	0.002	-0.435	0.473
152	0	0	0		152	0.233	-0.443	0.022	152	0.005	0.57	-1.562	152	0.009	-0.235	0.136
17	-0.844	-1.772	-0.104		17	0	0	0	17	0.001	-0.585	-0.429	17	-0.002	0.563	1.638
188	-1.198	-2.603	0.032		188	0	0	0	188	-0.002	0.052	-0.454	188	0.001	0.052	0.473
20	-0.832	-1.804	0.099		20	0	0	0	20	0.001	0.533	-1.621	20	0.001	-0.614	0.45
18	-2.307	1.772	0.106		18	0	0	0	18	-0.002	-0.585	-0.429	18	0.002	0.563	1.637
189	-4.198	2.604	-0.044		189	0	0	0	189	0.002	0.053	-0.454	189	-0.001	0.051	0.473
19	-2.342	1.803	-0.089		19	0	0	0	19	-0.001	0.533	-1.62	19	-0.001	-0.614	0.45
68	0	0	0		68	0.531	0.12	0.025	68	0.013	-0.294	-0.064	68	-0.01	0.091	0.019
69	0	0	0		69	0.563	-0.023	0.026	69	0.015	0.3	-0.065	69	0.001	-0.09	0.02
72	0	0	0		72	0.033	0.1	-0.005	72	-0.037	0.188	-0.933	72	0.005	-0.394	0.167
73	0	0	0		73	0.032	-0.106	0.012	73	-0.034	0.267	-0.849	73	0.006	-0.452	0.168

FOUNDATION REACTIONS 3

FOR NODE LOCATIONS REFER DRAWING NO: SASL-RS-DWG-ST-20K052-000105

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	12/10/2020	FIRST ISSUE	---	SGC	---	PRI	

SIEMENS Ingenuity for life

DRAWING NUMBER					REVISION
SASL-RS-DWG-ST-20K052-000108					1
DATE	28/09/2020	DATE	28/09/2020	DATE	28/09/2020
DRAWN	VG	CHECKED	FS	APPROVED	SJT

Outotec
The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

LARSEN & TOUBRO
This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER	REV
019045_008_S_G_26162_010	0
OUTOTEC DRAWING NUMBER: 900707_GAU01_WG01_MTL01_45137	DOCUMENT ID: 0U100703349 — REV 0

DRAWN	---	---
CHECKED	SGC	12/10/2020
ENGINEER	---	---
APPROVED	PRI	12/10/2020
EMPLOYER		
SCALE		A3

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
POWER GENERATION : POWER PLANT SUBSTATION
FOUNDATION LOAD DATA

DRAWING NUMBER	REV
MD-216-49PR-EG-ST-GAD-V514-2010	A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C02	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE OF FOUNDATION MKD. F2

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    OCT 12, 2020                *
*           Time=    14:59:41                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\SHARE\MAADEN\DG POWER PLANT\SUBSTATION\F2.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 -1; 14 0 -4 -1.5; 16 0 -4 -0.05; 24 0.5 -4 -0.05; 25 0.5 -4 -0.3
     9. 26 0.25 -4 -0.05; 27 0.25 -4 -0.3; 28 0 -4 -0.3; 31 0.5 -4 -0.5
    10. 32 0.25 -4 -0.5; 33 0 -4 -0.5; 36 0.5 -4 -0.75; 37 0.25 -4 -0.75
    11. 38 0 -4 -0.75; 41 0.5 -4 -1; 42 0.25 -4 -1; 45 0.5 -4 -1.25; 46 0.25 -4 -1.25
    12. 47 0 -4 -1.25; 70 0.5 -4 -1.5; 71 0.25 -4 -1.5; 74 0.5 -4 -1.75
    13. 75 0.25 -4 -1.75; 76 0 -4 -1.75; 79 0.5 -4 -2; 80 0.25 -4 -2; 81 0 -4 -2
    14. 84 0.5 -4 -2.25; 85 0.25 -4 -2.25; 86 0 -4 -2.25; 90 -0.25 -4 -0.05
    15. 91 -0.25 -4 -0.3; 92 -0.5 -4 -0.05; 93 -0.5 -4 -0.3; 97 -0.25 -4 -0.5
    16. 98 -0.5 -4 -0.5; 101 -0.25 -4 -0.75; 102 -0.5 -4 -0.75; 105 -0.25 -4 -1.
    17. 106 -0.5 -4 -1; 109 -0.25 -4 -1.25; 110 -0.5 -4 -1.25; 129 -0.25 -4 -1.5
    18. 130 -0.5 -4 -1.5; 133 -0.25 -4 -1.75; 134 -0.5 -4 -1.75; 137 -0.25 -4 -2
    19. 138 -0.5 -4 -2; 141 -0.25 -4 -2.25; 142 -0.5 -4 -2.25; 145 0 0.5 -2.5
    20. 146 0 0.5 0.2; 149 0.5 -4 -2.5; 150 0.25 -4 -2.5; 151 0 -4 -2.5
    21. 152 -0.25 -4 -2.5; 153 -0.5 -4 -2.5; 158 0.5 -4 0.2; 159 0.25 -4 0.2
    22. 160 0 -4 0.2; 161 -0.25 -4 0.2; 162 -0.5 -4 0.2; 164 0.5 -4 -2.75
    23. 166 0.25 -4 -2.75; 167 0 -4 -2.75; 168 -0.25 -4 -2.75; 169 -0.5 -4 -2.75
    24. 171 0.5 -4 -3; 173 0.25 -4 -3; 174 0 -4 -3; 175 -0.25 -4 -3; 176 -0.5 -4 -3
    25. 179 0.5 -4 0.45; 180 0.25 -4 0.45; 181 0 -4 0.45; 182 -0.25 -4 0.45
    26. 183 -0.5 -4 0.45; 186 0.5 -4 0.7; 187 0.25 -4 0.7; 188 0 -4 0.7
    27. 189 -0.25 -4 0.7; 190 -0.5 -4 0.7
    28. MEMBER INCIDENCES
    29. 133 151 145; 134 160 146
    30. ELEMENT INCIDENCES SHELL
    31. 27 24 26 27 25; 28 26 16 28 27; 31 25 27 32 31; 32 27 28 33 32; 35 31 32 37 36
    32. 36 32 33 38 37; 39 36 37 42 41; 40 37 38 13 42; 43 41 42 46 45; 44 42 13 47 46
    33. 63 45 46 71 70; 64 46 47 14 71; 67 70 71 75 74; 68 71 14 76 75; 71 74 75 80 79
    34. 72 75 76 81 80; 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91
    35. 85 28 91 97 33; 86 91 93 98 97; 89 33 97 101 38; 90 97 98 102 101
    36. 93 38 101 105 13; 94 101 102 106 105; 97 13 105 109 47; 98 105 106 110 109
    37. 117 47 109 129 14; 118 109 110 130 129; 121 14 129 133 76; 122 129 130 134 133
    38. 125 76 133 137 81; 126 133 134 138 137; 129 81 137 141 86; 130 137 138 142 141
```

```
39. 137 84 85 150 149; 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152
40. 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90
41. 151 149 150 166 164; 152 150 151 167 166; 153 151 152 168 167
42. 154 152 153 169 168; 157 164 166 173 171; 158 166 167 174 173
43. 159 167 168 175 174; 160 168 169 176 175; 163 179 180 159 158
44. 164 180 181 160 159; 165 181 182 161 160; 166 182 183 162 161
45. 169 186 187 180 179; 170 187 188 181 180; 171 188 189 182 181
46. 172 189 190 183 182
47. START GROUP DEFINITION
48. ELEMENT
49. _MOMENT 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 -
50. 89 90 93 94 97 98 117 118 121 122 125 126 129 130 137 140 145 148 151 154 -
51. 157 TO 160 163 166 169 TO 172
52. _SHEAR 31 32 35 36 39 40 43 44 63 64 67 68 71 72 85 86 89 90 93 94 97 98 117 -
53. 118 121 122 125 126
54. END GROUP DEFINITION
55. ELEMENT PROPERTY
56. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
57. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
58. 157 TO 160 163 TO 166 169 TO 172 THICKNESS 0.3
59. SUPPORTS
60. 13 14 16 24 TO 28 31 TO 33 36 TO 38 41 42 45 TO 47 70 71 74 TO 76 79 TO 81 -
61. 84 TO 86 90 TO 93 97 98 101 102 105 106 109 110 129 130 133 134 137 138 141 -
62. 142 149 TO 153 158 TO 162 164 166 TO 169 171 173 TO 176 179 TO 183 -
63. 186 TO 190 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
```

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.062500
24	0.031250
25	0.028125
26	0.062500
27	0.056250
28	0.056250
31	0.028125
32	0.056250
33	0.056250
36	0.031250
37	0.062500
38	0.062500
41	0.031250
42	0.062500
45	0.031250
46	0.062500
47	0.062500
70	0.031250
71	0.062500
74	0.031250
75	0.062500
76	0.062500
79	0.031250
80	0.062500
81	0.062500

84	0.031250
85	0.062500
86	0.062500
90	0.062500
91	0.056250
92	0.031250
93	0.028125
97	0.056250
98	0.028125
101	0.062500
102	0.031250
105	0.062500
106	0.031250
109	0.062500
110	0.031250
129	0.062500
130	0.031250
133	0.062500
134	0.031250
137	0.062500
138	0.031250
141	0.062500
142	0.031250
149	0.031250
150	0.062500
151	0.062500
152	0.062500
153	0.031250
158	0.031250
159	0.062500
160	0.062500
161	0.062500
162	0.031250
164	0.031250
166	0.062500
167	0.062500
168	0.062500
169	0.031250
171	0.015625
173	0.031250
174	0.031250
175	0.031250
176	0.015625
179	0.031250
180	0.062500
181	0.062500
182	0.062500
183	0.031250
186	0.015625
187	0.031250
188	0.031250
189	0.031250
190	0.015625
TOTAL	3.700000

64. DEFINE MATERIAL START
65. ISOTROPIC STEEL

```
 66. E 2.09042E+007
 67. POISSON 0.3
 68. DENSITY 7.83341
 69. ALPHA 1.2E-005
 70. DAMP 0.03
 71. TYPE STEEL
 72. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
 73. ISOTROPIC CONCRETE
 74. E 2.21467E+006
 75. POISSON 0.17
 76. DENSITY 2.40262
 77. ALPHA 1E-005
 78. DAMP 0.05
 79. TYPE CONCRETE
 80. STRENGTH FCU 2812.28
 81. END DEFINE MATERIAL
 82. MEMBER PROPERTY AMERICAN
 83. 133 134 PRIS YD 0.6 ZD 0.6
 84. CONSTANTS
 85. MATERIAL CONCRETE ALL
 86. *****ALLOWABLE STRESS DESIGN LOAD COMBS*********************************
 87. LOAD 1 DL+LR
 88. SELFWEIGHT Y -1
 89. ELEMENT LOAD
 90. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
 91. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
 92. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
 93. JOINT LOAD
 94. 145 FY -14 FZ -0.1
 95. 146 FY -14 FZ -0.1
 96. ***************************************************
 97. LOAD 2 DL+0.75*LR+0.75*WL1
 98. JOINT LOAD
 99. 145 FX 0 FY -11 FZ -0.1
100. 146 FX 0 FY -11 FZ -0.1
101. SELFWEIGHT Y -1
102. ELEMENT LOAD
103. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
104. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
105. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
106. ***************************************************
107. LOAD 3 DL+0.75*LR+0.75*WL2
108. SELFWEIGHT Y -1
109. ELEMENT LOAD
110. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
111. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
112. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
113. JOINT LOAD
114. 145 FX 0 FY -11 FZ -0.1
115. 146 FX 0 FY -11 FZ -0.1
116. ***************************************************
117. LOAD 4 DL+0.75*LR+0.75*WL3
118. SELFWEIGHT Y -1
119. ELEMENT LOAD
120. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
121. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
```

```
122. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
123. JOINT LOAD
124. 145 FX 0 FY -11 FZ -0.2
125. 146 FX 0 FY -11 FZ -0.2
126. ******************************************************
127. LOAD 5 DL+0.75*LR+0.75*WL4
128. SELFWEIGHT Y -1
129. ELEMENT LOAD
130. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
131. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
132. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
133. JOINT LOAD
134. 145 FX 0 FY -11 FZ -0.1
135. 146 FX 0 FY -11 FZ -0.1
136. ******************************************************
137. LOAD 6 0.6*DL+WL1
138. SELFWEIGHT Y -1
139. ELEMENT LOAD
140. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
141. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
142. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
143. JOINT LOAD
144. 145 FX 0 FY -0.9 FZ -0.1
145. 146 FX 0 FY -0.9 FZ -0.1
146. ******************************************************
147. LOAD 7 0.6*DL+WL2
148. SELFWEIGHT Y -1
149. ELEMENT LOAD
150. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
151. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
152. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
153. JOINT LOAD
154. 145 FX 0 FY -0.9 FZ -0.1
155. 146 FX 0 FY -0.9 FZ -0.1
156. ******************************************************
157. LOAD 8 0.6*DL+WL3
158. SELFWEIGHT Y -1
159. ELEMENT LOAD
160. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
161. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
162. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
163. JOINT LOAD
164. 145 FX 0 FY -0.9 FZ -0.1
165. 146 FX 0 FY -0.9 FZ -0.1
166. ******************************************************
167. LOAD 9 0.6*DL+WL4
168. SELFWEIGHT Y -1
169. ELEMENT LOAD
170. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
171. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
172. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
173. JOINT LOAD
174. 145 FX 0 FY -0.9 FZ -0.1
175. 146 FX 0 FY -0.9 FZ -0.1
176. ******************************************************************************
177. LOAD 10 DL+0.75*LR+0.75*0.7*EL1
```

```
178. JOINT LOAD
179. 145 FX 0 FY -11 FZ -0.1
180. 146 FX 0 FY -11 FZ -0.1
181. SELFWEIGHT Y -1
182. ELEMENT LOAD
183. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
184. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
185. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
186. ************************************************************
187. LOAD 11 DL+0.75*LR+0.75*0.7*EL2
188. SELFWEIGHT Y -1
189. ELEMENT LOAD
190. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
191. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
192. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
193. JOINT LOAD
194. 145 FX 0 FY -11 FZ -0.1
195. 146 FX 0 FY -11 FZ -0.1
196. ************************************************************
197. LOAD 12 DL+0.75*LR+0.75*0.7*EL3
198. SELFWEIGHT Y -1
199. ELEMENT LOAD
200. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
201. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
202. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
203. JOINT LOAD
204. 145 FX 0 FY -11 FZ -0.1
205. 146 FX 0 FY -11 FZ -0.1
206. ************************************************************
207. LOAD 13 DL+0.75*LR+0.75*0.7*EL4
208. SELFWEIGHT Y -1
209. ELEMENT LOAD
210. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
211. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
212. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
213. JOINT LOAD
214. 145 FX 0 FY -11 FZ -0.1
215. 146 FX 0 FY -11 FZ -0.1
216. ************************************************************
217. LOAD 14 0.6*DL+0.7*EL1
218. SELFWEIGHT Y -1
219. ELEMENT LOAD
220. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
221. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
222. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
223. JOINT LOAD
224. 145 FX 0 FY -0.9 FZ 0
225. 146 FX 0 FY -0.9 FZ 0
226. ************************************************************
227. LOAD 15 0.6*DL+0.7*EL2
228. SELFWEIGHT Y -1
229. ELEMENT LOAD
230. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
231. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
232. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
233. JOINT LOAD
```

```
234. 145 FX 0 FY -0.9 FZ 0
235. 146 FX 0 FY -0.9 FZ 0
236. ************************************************
237. LOAD 16 0.6*DL+0.7*EL3
238. SELFWEIGHT Y -1
239. ELEMENT LOAD
240. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
241. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
242. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
243. JOINT LOAD
244. 145 FX 0 FY -0.9 FZ 0
245. 146 FX 0 FY -0.9 FZ 0
246. ************************************************
247. LOAD 17 0.6*DL+0.7*EL4
248. SELFWEIGHT Y -1
249. ELEMENT LOAD
250. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
251. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
252. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
253. JOINT LOAD
254. 145 FX 0 FY -0.9 FZ 0
255. 146 FX 0 FY -0.9 FZ 0
256. ********************************************************************************
257. *****ULTIMATE STRENGTH DESIGN LOAD COMBS*************************************
258. LOAD 100 1.4*DL
259. SELFWEIGHT Y -1
260. ELEMENT LOAD
261. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
262. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
263. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.6
264. JOINT LOAD
265. 145 FX 0 FY -2 FZ 0
266. 146 FX 0 FY -2 FZ 0
267. ************************************************
268. LOAD 101 1.4D+1.7(LL)
269. SELFWEIGHT Y -1
270. ELEMENT LOAD
271. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
272. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
273. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.6
274. JOINT LOAD
275. 145 FX 0 FY -23 FZ -0.2
276. 146 FX 0 FY -23 FZ -0.2
277. ************************************************
278. LOAD 102 1.2D+1.6WL1+0.5L
279. SELFWEIGHT Y -1
280. ELEMENT LOAD
281. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
282. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
283. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
284. JOINT LOAD
285. 145 FX 0 FY -8 FZ -0.2
286. 146 FX 0 FY -8 FZ -0.2
287. ************************************************
288. LOAD 103 1.2D+1.6WL2+0.5L
289. SELFWEIGHT Y -1
```

```
290. ELEMENT LOAD
291. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
292. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
293. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
294. JOINT LOAD
295. 145 FX 0 FY -8 FZ -0.2
296. 146 FX 0 FY -8 FZ -0.2
297. ************************************************
298. LOAD 104 1.2D+1.6WL3+0.5L
299. SELFWEIGHT Y -1
300. ELEMENT LOAD
301. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
302. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
303. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
304. JOINT LOAD
305. 145 FX 0 FY -8 FZ -0.2
306. 146 FX 0 FY -8 FZ -0.2
307. ************************************************8
308. LOAD 105 1.2D+1.6WL4+0.5L
309. SELFWEIGHT Y -1
310. ELEMENT LOAD
311. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
312. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
313. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
314. JOINT LOAD
315. 145 FX 0 FY -8 FZ -0.2
316. 146 FX 0 FY -8 FZ -0.2
317. ************************************************************************
318. LOAD 106 0.9D+1.6WL1
319. SELFWEIGHT Y -1
320. ELEMENT LOAD
321. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
322. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
323. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
324. JOINT LOAD
325. 145 FX 0 FY -1.4 FZ -0.1
326. 146 FX 0 FY -1.4 FZ -0.1
327. ************************************************************************
328. LOAD 107 0.9D+1.6WL2
329. SELFWEIGHT Y -1
330. ELEMENT LOAD
331. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
332. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
333. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
334. JOINT LOAD
335. 145 FX 0 FY -1.4 FZ -0.1
336. 146 FX 0 FY -1.4 FZ -0.1
337. ************************************************************************
338. LOAD 108 0.9D+1.6WL3
339. SELFWEIGHT Y -1
340. ELEMENT LOAD
341. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
342. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
343. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
344. JOINT LOAD
345. 145 FX 0 FY -1.4 FZ -0.2
```

```
346. 146 FX 0 FY -1.4 FZ -0.2
347. ************************************************************************
348. LOAD 109 0.9D+1.6WL4
349. SELFWEIGHT Y -1
350. ELEMENT LOAD
351. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
352. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
353. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
354. JOINT LOAD
355. 145 FX 0 FY -1.4 FZ -0.1
356. 146 FX 0 FY -1.4 FZ -0.1
357. ************************************************************************
358. LOAD 110 1.2D+1.0EL1+0.5L
359. SELFWEIGHT Y -1
360. ELEMENT LOAD
361. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
362. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
363. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
364. JOINT LOAD
365. 145 FX 0 FY -8 FZ -0.2
366. 146 FX 0 FY -8 FZ -0.2
367. ************************************************
368. LOAD 111 1.2D+1.0EL2+0.5L
369. SELFWEIGHT Y -1
370. ELEMENT LOAD
371. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
372. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
373. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
374. JOINT LOAD
375. 145 FX 0 FY -8 FZ -0.2
376. 146 FX 0 FY -8 FZ -0.2
377. ************************************************
378. LOAD 112 1.2D+1.0EL3+0.5L
379. SELFWEIGHT Y -1
380. ELEMENT LOAD
381. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
382. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
383. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
384. JOINT LOAD
385. 145 FX 0 FY -8 FZ -0.2
386. 146 FX 0 FY -8 FZ -0.2
387. ************************************************8
388. LOAD 113 1.2D+1.0EL4+0.5L
389. SELFWEIGHT Y -1
390. ELEMENT LOAD
391. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
392. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
393. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
394. JOINT LOAD
395. 145 FX 0 FY -8 FZ -0.2
396. 146 FX 0 FY -8 FZ -0.2
397. ************************************************************************
398. LOAD 114 0.9D+1.0EL1
399. SELFWEIGHT Y -1
400. ELEMENT LOAD
401. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
```

```
402. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
403. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
404. JOINT LOAD
405. 145 FX 0 FY -1.4 FZ -0.1
406. 146 FX 0 FY -1.4 FZ -0.1
407. ******************************************************************************
408. LOAD 115 0.9D+1.0EL2
409. SELFWEIGHT Y -1
410. ELEMENT LOAD
411. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
412. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
413. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
414. JOINT LOAD
415. 145 FX 0 FY -1.4 FZ -0.1
416. 146 FX 0 FY -1.4 FZ -0.1
417. ******************************************************************************
418. LOAD 116 0.9D+1.0EL3
419. SELFWEIGHT Y -1
420. ELEMENT LOAD
421. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
422. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
423. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
424. JOINT LOAD
425. 145 FX 0 FY -1.4 FZ -0.2
426. 146 FX 0 FY -1.4 FZ -0.2
427. ******************************************************************************
428. LOAD 117 0.9D+1.0EL4
429. SELFWEIGHT Y -1
430. ELEMENT LOAD
431. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
432. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
433. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
434. JOINT LOAD
435. 145 FX 0 FY -1.4 FZ -0.1
436. 146 FX 0 FY -1.4 FZ -0.1
437. ******************************************************************************
438. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**82**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**60**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**80**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 58/ 10/ 33 DOF
TOTAL PRIMARY LOAD CASES = 35, TOTAL DEGREES OF FREEDOM = 252
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 9 DOUBLE KILO-WORDS

REQRD/AVAIL. DISK SPACE = 12.5/ 160016.0 MB

```
439. LOAD LIST 100 TO 117
440. START CONCRETE DESIGN
```

```
441. CODE ACI 1999
442. TRACK 0 ALL
443. MAXMAIN 25 ALL
444. CLB 0.075 MEMB 133 134
445. CLS 0.075 ALL
446. DESIGN COLUMN 133 134
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

447. END CONCRETE DESIGN
448. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 12,2020 TIME= 14:59:47 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE OF FOUNDATION MKD. F4

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    NOV  4, 2020                *
*           Time=    10:14: 1                    *
*                                                *
*      USER ID:                                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: E:\SHARE\MAADEN\DG POWER PLANT\SUBSTATION\F4.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 -1; 14 0 -4 -1.5; 16 0 -4 -0.05; 24 0.5 -4 -0.05; 25 0.5 -4 -0.3
     9. 26 0.25 -4 -0.05; 27 0.25 -4 -0.3; 28 0 -4 -0.3; 31 0.5 -4 -0.5
    10. 32 0.25 -4 -0.5; 33 0 -4 -0.5; 36 0.5 -4 -0.75; 37 0.25 -4 -0.75
    11. 38 0 -4 -0.75; 41 0.5 -4 -1; 42 0.25 -4 -1; 45 0.5 -4 -1.25; 46 0.25 -4 -1.25
    12. 47 0 -4 -1.25; 70 0.5 -4 -1.5; 71 0.25 -4 -1.5; 74 0.5 -4 -1.75
    13. 75 0.25 -4 -1.75; 76 0 -4 -1.75; 79 0.5 -4 -2; 80 0.25 -4 -2; 81 0 -4 -2
    14. 84 0.5 -4 -2.25; 85 0.25 -4 -2.25; 86 0 -4 -2.25; 90 -0.25 -4 -0.05
    15. 91 -0.25 -4 -0.3; 92 -0.5 -4 -0.05; 93 -0.5 -4 -0.3; 97 -0.25 -4 -0.5
    16. 98 -0.5 -4 -0.5; 101 -0.25 -4 -0.75; 102 -0.5 -4 -0.75; 105 -0.25 -4 -1.
    17. 106 -0.5 -4 -1; 109 -0.25 -4 -1.25; 110 -0.5 -4 -1.25; 129 -0.25 -4 -1.5
    18. 130 -0.5 -4 -1.5; 133 -0.25 -4 -1.75; 134 -0.5 -4 -1.75; 137 -0.25 -4 -2
    19. 138 -0.5 -4 -2; 141 -0.25 -4 -2.25; 142 -0.5 -4 -2.25; 145 0 -2.4 -2.5
    20. 146 0 -2.4 0.25; 149 0.5 -4 -2.5; 150 0.25 -4 -2.5; 151 0 -4 -2.5
    21. 152 -0.25 -4 -2.5; 153 -0.5 -4 -2.5; 158 0.5 -4 0.25; 159 0.25 -4 0.25
    22. 160 0 -4 0.25; 161 -0.25 -4 0.25; 162 -0.5 -4 0.25; 164 0.5 -4 -2.75
    23. 166 0.25 -4 -2.75; 167 0 -4 -2.75; 168 -0.25 -4 -2.75; 169 -0.5 -4 -2.75
    24. 171 0.5 -4 -3; 173 0.25 -4 -3; 174 0 -4 -3; 175 -0.25 -4 -3; 176 -0.5 -4 -3
    25. 179 0.5 -4 0.5; 180 0.25 -4 0.5; 181 0 -4 0.5; 182 -0.25 -4 0.5
    26. 183 -0.5 -4 0.5; 186 0.5 -4 0.75; 187 0.25 -4 0.75; 188 0 -4 0.75
    27. 189 -0.25 -4 0.75; 190 -0.5 -4 0.75
    28. MEMBER INCIDENCES
    29. 133 151 145; 134 160 146
    30. ELEMENT INCIDENCES SHELL
    31. 27 24 26 27 25; 28 26 16 28 27; 31 25 27 32 31; 32 27 28 33 32; 35 31 32 37 36
    32. 36 32 33 38 37; 39 36 37 42 41; 40 37 38 13 42; 43 41 42 46 45; 44 42 13 47 46
    33. 63 45 46 71 70; 64 46 47 14 71; 67 70 71 75 74; 68 71 14 76 75; 71 74 75 80 79
    34. 72 75 76 81 80; 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91
    35. 85 28 91 97 33; 86 91 93 98 97; 89 33 97 101 38; 90 97 98 102 101
    36. 93 38 101 105 13; 94 101 102 106 105; 97 13 105 109 47; 98 105 106 110 109
    37. 117 47 109 129 14; 118 109 110 130 129; 121 14 129 133 76; 122 129 130 134 133
    38. 125 76 133 137 81; 126 133 134 138 137; 129 81 137 141 86; 130 137 138 142 141
```

```
39. 137 84 85 150 149; 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152
40. 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90
41. 151 149 150 166 164; 152 150 151 167 166; 153 151 152 168 167
42. 154 152 153 169 168; 157 164 166 173 171; 158 166 167 174 173
43. 159 167 168 175 174; 160 168 169 176 175; 163 179 180 159 158
44. 164 180 181 160 159; 165 181 182 161 160; 166 182 183 162 161
45. 169 186 187 180 179; 170 187 188 181 180; 171 188 189 182 181
46. 172 189 190 183 182
47. START GROUP DEFINITION
48. ELEMENT
49. _MOMENT 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 -
50. 89 90 93 94 97 98 117 118 121 122 125 126 129 130 137 140 145 148 151 154 -
51. 157 TO 160 163 166 169 TO 172
52. _SHEAR 31 32 35 36 39 40 43 44 63 64 67 68 71 72 85 86 89 90 93 94 97 98 117 -
53. 118 121 122 125 126
54. END GROUP DEFINITION
55. ELEMENT PROPERTY
56. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
57. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
58. 157 TO 160 163 TO 166 169 TO 172 THICKNESS 0.3
59. SUPPORTS
60. 13 14 16 24 TO 28 31 TO 33 36 TO 38 41 42 45 TO 47 70 71 74 TO 76 79 TO 81 -
61. 84 TO 86 90 TO 93 97 98 101 102 105 106 109 110 129 130 133 134 137 138 141 -
62. 142 149 TO 153 158 TO 162 164 166 TO 169 171 173 TO 176 179 TO 183 -
63. 186 TO 190 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
```

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.068750
24	0.034375
25	0.028125
26	0.068750
27	0.056250
28	0.056250
31	0.028125
32	0.056250
33	0.056250
36	0.031250
37	0.062500
38	0.062500
41	0.031250
42	0.062500
45	0.031250
46	0.062500
47	0.062500
70	0.031250
71	0.062500
74	0.031250
75	0.062500
76	0.062500
79	0.031250
80	0.062500
81	0.062500

84	0.031250
85	0.062500
86	0.062500
90	0.068750
91	0.056250
92	0.034375
93	0.028125
97	0.056250
98	0.028125
101	0.062500
102	0.031250
105	0.062500
106	0.031250
109	0.062500
110	0.031250
129	0.062500
130	0.031250
133	0.062500
134	0.031250
137	0.062500
138	0.031250
141	0.062500
142	0.031250
149	0.031250
150	0.062500
151	0.062500
152	0.062500
153	0.031250
158	0.034375
159	0.068750
160	0.068750
161	0.068750
162	0.034375
164	0.031250
166	0.062500
167	0.062500
168	0.062500
169	0.031250
171	0.015625
173	0.031250
174	0.031250
175	0.031250
176	0.015625
179	0.031250
180	0.062500
181	0.062500
182	0.062500
183	0.031250
186	0.015625
187	0.031250
188	0.031250
189	0.031250
190	0.015625
TOTAL	3.750000

64. DEFINE MATERIAL START
65. ISOTROPIC STEEL

```
66. E 2.09042E+007
67. POISSON 0.3
68. DENSITY 7.83341
69. ALPHA 1.2E-005
70. DAMP 0.03
71. TYPE STEEL
72. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
73. ISOTROPIC CONCRETE
74. E 2.21467E+006
75. POISSON 0.17
76. DENSITY 2.40262
77. ALPHA 1E-005
78. DAMP 0.05
79. TYPE CONCRETE
80. STRENGTH FCU 2812.28
81. END DEFINE MATERIAL
82. MEMBER PROPERTY AMERICAN
83. 133 134 PRIS YD 0.5 ZD 0.325
84. CONSTANTS
85. MATERIAL CONCRETE ALL
86. *****ALLOWABLE STRESS DESIGN LOAD COMBS*********************************
87. LOAD 1 DL+LR
88. SELFWEIGHT Y -1
89. ELEMENT LOAD
90. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
91. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
92. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
93. JOINT LOAD
94. 145 FY -3.5 FZ -0.1
95. 146 FY -3.5 FZ -0.1
96. ***************************************************
97. LOAD 2 DL+0.75*LR+0.75*WL1
98. JOINT LOAD
99. 145 FX 0 FY -3 FZ -0.1
100. 146 FX 0 FY -3 FZ -0.1
101. SELFWEIGHT Y -1
102. ELEMENT LOAD
103. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
104. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
105. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
106. ***************************************************
107. LOAD 3 DL+0.75*LR+0.75*WL2
108. SELFWEIGHT Y -1
109. ELEMENT LOAD
110. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
111. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
112. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
113. JOINT LOAD
114. 145 FX 0 FY -3 FZ -0.1
115. 146 FX 0 FY -3 FZ -0.1
116. ***************************************************
117. LOAD 4 DL+0.75*LR+0.75*WL3
118. SELFWEIGHT Y -1
119. ELEMENT LOAD
120. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
121. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
```

```
122. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
123. JOINT LOAD
124. 145 FX 0 FY -3 FZ -0.2
125. 146 FX 0 FY -3 FZ -0.2
126. ******************************************************
127. LOAD 5 DL+0.75*LR+0.75*WL4
128. SELFWEIGHT Y -1
129. ELEMENT LOAD
130. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
131. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
132. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
133. JOINT LOAD
134. 145 FX 0 FY -3 FZ -0.1
135. 146 FX 0 FY -3 FZ -0.1
136. ******************************************************
137. LOAD 6 0.6*DL+WL1
138. SELFWEIGHT Y -1
139. ELEMENT LOAD
140. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
141. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
142. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
143. JOINT LOAD
144. 145 FX 0 FY -0.9 FZ -0.1
145. 146 FX 0 FY -0.9 FZ -0.1
146. ******************************************************
147. LOAD 7 0.6*DL+WL2
148. SELFWEIGHT Y -1
149. ELEMENT LOAD
150. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
151. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
152. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
153. JOINT LOAD
154. 145 FX 0 FY -0.9 FZ -0.1
155. 146 FX 0 FY -0.9 FZ -0.1
156. ******************************************************
157. LOAD 8 0.6*DL+WL3
158. SELFWEIGHT Y -1
159. ELEMENT LOAD
160. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
161. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
162. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
163. JOINT LOAD
164. 145 FX 0 FY -0.9 FZ -0.1
165. 146 FX 0 FY -0.9 FZ -0.1
166. ******************************************************
167. LOAD 9 0.6*DL+WL4
168. SELFWEIGHT Y -1
169. ELEMENT LOAD
170. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
171. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
172. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
173. JOINT LOAD
174. 145 FX 0 FY -0.9 FZ -0.1
175. 146 FX 0 FY -0.9 FZ -0.1
176. ******************************************************************************
177. LOAD 10 DL+0.75*LR+0.75*0.7*EL1
```

```
178. JOINT LOAD
179. 145 FX 0 FY -3 FZ -0.1
180. 146 FX 0 FY -3 FZ -0.1
181. SELFWEIGHT Y -1
182. ELEMENT LOAD
183. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
184. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
185. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
186. ************************************************************
187. LOAD 11 DL+0.75*LR+0.75*0.7*EL2
188. SELFWEIGHT Y -1
189. ELEMENT LOAD
190. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
191. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
192. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
193. JOINT LOAD
194. 145 FX 0 FY -3 FZ -0.1
195. 146 FX 0 FY -3 FZ -0.1
196. ************************************************************
197. LOAD 12 DL+0.75*LR+0.75*0.7*EL3
198. SELFWEIGHT Y -1
199. ELEMENT LOAD
200. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
201. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
202. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
203. JOINT LOAD
204. 145 FX 0 FY -3 FZ -0.1
205. 146 FX 0 FY -3 FZ -0.1
206. ************************************************************
207. LOAD 13 DL+0.75*LR+0.75*0.7*EL4
208. SELFWEIGHT Y -1
209. ELEMENT LOAD
210. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
211. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
212. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.2
213. JOINT LOAD
214. 145 FX 0 FY -3 FZ -0.1
215. 146 FX 0 FY -3 FZ -0.1
216. ************************************************************
217. LOAD 14 0.6*DL+0.7*EL1
218. SELFWEIGHT Y -1
219. ELEMENT LOAD
220. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
221. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
222. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
223. JOINT LOAD
224. 145 FX 0 FY -0.9 FZ 0
225. 146 FX 0 FY -0.9 FZ 0
226. ************************************************************
227. LOAD 15 0.6*DL+0.7*EL2
228. SELFWEIGHT Y -1
229. ELEMENT LOAD
230. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
231. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
232. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
233. JOINT LOAD
```

```
234. 145 FX 0 FY -0.9 FZ 0
235. 146 FX 0 FY -0.9 FZ 0
236. ************************************************
237. LOAD 16 0.6*DL+0.7*EL3
238. SELFWEIGHT Y -1
239. ELEMENT LOAD
240. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
241. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
242. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
243. JOINT LOAD
244. 145 FX 0 FY -0.9 FZ 0
245. 146 FX 0 FY -0.9 FZ 0
246. ************************************************
247. LOAD 17 0.6*DL+0.7*EL4
248. SELFWEIGHT Y -1
249. ELEMENT LOAD
250. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
251. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
252. 157 TO 160 163 TO 166 169 TO 172 PR GY -2.8
253. JOINT LOAD
254. 145 FX 0 FY -0.9 FZ 0
255. 146 FX 0 FY -0.9 FZ 0
256. ********************************************************************************
257. *****ULTIMATE STRENGTH DESIGN LOAD COMBS*************************************
258. LOAD 100 1.4*DL
259. SELFWEIGHT Y -1
260. ELEMENT LOAD
261. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
262. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
263. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.6
264. JOINT LOAD
265. 145 FX 0 FY -1 FZ 0
266. 146 FX 0 FY -1 FZ 0
267. ************************************************
268. LOAD 101 1.4D+1.7(LL)
269. SELFWEIGHT Y -1
270. ELEMENT LOAD
271. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
272. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
273. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.6
274. JOINT LOAD
275. 145 FX 0 FY -6 FZ -0.2
276. 146 FX 0 FY -6 FZ -0.2
277. ************************************************
278. LOAD 102 1.2D+1.6WL1+0.5L
279. SELFWEIGHT Y -1
280. ELEMENT LOAD
281. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
282. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
283. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
284. JOINT LOAD
285. 145 FX 0 FY -2.2 FZ -0.2
286. 146 FX 0 FY -2.2 FZ -0.2
287. ************************************************
288. LOAD 103 1.2D+1.6WL2+0.5L
289. SELFWEIGHT Y -1
```

```
ELEMENT LOAD
27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
JOINT LOAD
145 FX 0 FY -2.2 FZ -0.2
146 FX 0 FY -2.2 FZ -0.2
**********************************************
LOAD 104 1.2D+1.6WL3+0.5L
SELFWEIGHT Y -1
ELEMENT LOAD
27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
JOINT LOAD
145 FX 0 FY -2.2 FZ -0.2
146 FX 0 FY -2.2 FZ -0.2
**********************************************8
LOAD 105 1.2D+1.6WL4+0.5L
SELFWEIGHT Y -1
ELEMENT LOAD
27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
JOINT LOAD
145 FX 0 FY -2.2 FZ -0.2
146 FX 0 FY -2.2 FZ -0.2
**********************************************************************
LOAD 106 0.9D+1.6WL1
SELFWEIGHT Y -1
ELEMENT LOAD
27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
JOINT LOAD
145 FX 0 FY -1.4 FZ -0.1
146 FX 0 FY -1.4 FZ -0.1
**************************************************************************
LOAD 107 0.9D+1.6WL2
SELFWEIGHT Y -1
ELEMENT LOAD
27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
JOINT LOAD
145 FX 0 FY -1.4 FZ -0.1
146 FX 0 FY -1.4 FZ -0.1
**********************************************************************
LOAD 108 0.9D+1.6WL3
SELFWEIGHT Y -1
ELEMENT LOAD
27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
JOINT LOAD
145 FX 0 FY -1.4 FZ -0.2
```

```
346. 146 FX 0 FY -1.4 FZ -0.2
347. ************************************************************************
348. LOAD 109 0.9D+1.6WL4
349. SELFWEIGHT Y -1
350. ELEMENT LOAD
351. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
352. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
353. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
354. JOINT LOAD
355. 145 FX 0 FY -1.4 FZ -0.1
356. 146 FX 0 FY -1.4 FZ -0.1
357. ************************************************************************
358. LOAD 110 1.2D+1.0EL1+0.5L
359. SELFWEIGHT Y -1
360. ELEMENT LOAD
361. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
362. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
363. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
364. JOINT LOAD
365. 145 FX 0 FY -2.2 FZ -0.2
366. 146 FX 0 FY -2.2 FZ -0.2
367. ************************************************
368. LOAD 111 1.2D+1.0EL2+0.5L
369. SELFWEIGHT Y -1
370. ELEMENT LOAD
371. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
372. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
373. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
374. JOINT LOAD
375. 145 FX 0 FY -2.2 FZ -0.2
376. 146 FX 0 FY -2.2 FZ -0.2
377. ************************************************
378. LOAD 112 1.2D+1.0EL3+0.5L
379. SELFWEIGHT Y -1
380. ELEMENT LOAD
381. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
382. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
383. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
384. JOINT LOAD
385. 145 FX 0 FY -2.2 FZ -0.2
386. 146 FX 0 FY -2.2 FZ -0.2
387. ************************************************8
388. LOAD 113 1.2D+1.0EL4+0.5L
389. SELFWEIGHT Y -1
390. ELEMENT LOAD
391. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
392. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
393. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.4
394. JOINT LOAD
395. 145 FX 0 FY -2.2 FZ -0.2
396. 146 FX 0 FY -2.2 FZ -0.2
397. ************************************************************************
398. LOAD 114 0.9D+1.0EL1
399. SELFWEIGHT Y -1
400. ELEMENT LOAD
401. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
```

```
402. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
403. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
404. JOINT LOAD
405. 145 FX 0 FY -1.4 FZ -0.1
406. 146 FX 0 FY -1.4 FZ -0.1
407. ******************************************************************************
408. LOAD 115 0.9D+1.0EL2
409. SELFWEIGHT Y -1
410. ELEMENT LOAD
411. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
412. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
413. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
414. JOINT LOAD
415. 145 FX 0 FY -1.4 FZ -0.1
416. 146 FX 0 FY -1.4 FZ -0.1
417. ******************************************************************************
418. LOAD 116 0.9D+1.0EL3
419. SELFWEIGHT Y -1
420. ELEMENT LOAD
421. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
422. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
423. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
424. JOINT LOAD
425. 145 FX 0 FY -1.4 FZ -0.2
426. 146 FX 0 FY -1.4 FZ -0.2
427. ******************************************************************************
428. LOAD 117 0.9D+1.0EL4
429. SELFWEIGHT Y -1
430. ELEMENT LOAD
431. 27 28 31 32 35 36 39 40 43 44 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 -
432. 94 97 98 117 118 121 122 125 126 129 130 137 TO 140 145 TO 148 151 TO 154 -
433. 157 TO 160 163 TO 166 169 TO 172 PR GY -3.1
434. JOINT LOAD
435. 145 FX 0 FY -1.4 FZ -0.1
436. 146 FX 0 FY -1.4 FZ -0.1
437. ******************************************************************************
438. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**82**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**60**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**80**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 58/ 10/ 33 DOF
TOTAL PRIMARY LOAD CASES = 35, TOTAL DEGREES OF FREEDOM = 252
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 9 DOUBLE KILO-WORDS

REQRD/AVAIL. DISK SPACE = 12.5/ 271960.1 MB

```
439. LOAD LIST 100 TO 117
440. START CONCRETE DESIGN
```

```
441. CODE ACI 1999
442. TRACK 0 ALL
443. MAXMAIN 25 ALL
444. CLB 0.075 MEMB 133 134
445. CLS 0.075 ALL
446. FC 3500 MEMB 133 134
447. DESIGN COLUMN 133 134
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, RECT SIZE - 325.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.208	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 325.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, RECT SIZE - 325.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1625.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.208	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 325.00 MM

**********END OF COLUMN DESIGN RESULTS**********

448. END CONCRETE DESIGN
449. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 4,2020 TIME= 10:14:10 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	C02	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023			Date	12-10-2020	04-11-2020	26-11-2020	26-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	ISSUED FOR CONSTRUCTION & REVISED AS MARKED	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI		

C02

ATTACHMENT 4 STAAD INPUT & OUTPUT FILE FOR R C FRAME MARKED FR-1

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    NOV 18, 2020                *
*           Time=    12:48:12                    *
*                                                *
*      USER ID:                                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: E:\SHARE\MAADEN\DG POWER PLANT\SUBSTATION\FRAME.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 145 0 0.5 -2.575; 146 7.4 0.5 -2.575; 151 0 -4 -2.575; 160 7.4 -4 -2.575
     9. 191 0 -0.5 -2.575; 192 7.4 -0.5 -2.575; 193 3.7 -0.5 -2.575
    10. 194 3.7 0.5 -2.575
    11. MEMBER INCIDENCES
    12. 133 151 191; 134 160 192; 173 191 145; 174 192 146; 175 191 193; 176 193 192
    13. 177 193 194
    14. SUPPORTS
    15. 151 160 FIXED
    16. DEFINE MATERIAL START
    17. ISOTROPIC STEEL
    18. E 2.09042E+007
    19. POISSON 0.3
    20. DENSITY 7.83341
    21. ALPHA 1.2E-005
    22. DAMP 0.03
    23. TYPE STEEL
    24. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
    25. ISOTROPIC CONCRETE
    26. E 2.21467E+006
    27. POISSON 0.17
    28. DENSITY 2.40262
    29. ALPHA 1E-005
    30. DAMP 0.05
    31. TYPE CONCRETE
    32. STRENGTH FCU 2812.28
    33. END DEFINE MATERIAL
    34. MEMBER PROPERTY AMERICAN
    35. 133 134 173 174 177 PRIS YD 0.65 ZD 0.65
    36. MEMBER PROPERTY AMERICAN
    37. 175 176 PRIS YD 0.75 ZD 0.5
    38. CONSTANTS
```

```
39. MATERIAL CONCRETE ALL
40. *****ALLOWABLE STRESS DESIGN LOAD COMBS*********************************
41. LOAD 1 DL+LR
42. SELFWEIGHT Y -1
43. JOINT LOAD
44. 145 FY -9.4
45. 194 FY -11 FZ 1.3
46. 146 FY -12 FZ 2
47. ******************************************************
48. LOAD 2 DL+0.75*LR+0.75*WL1
49. JOINT LOAD
50. 145 FX -0.1 FY -8
51. 194 FX 0 FY -9 FZ 1.2
52. 146 FX 0 FY -10 FZ 1.75
53. SELFWEIGHT Y -1
54. ******************************************************
55. LOAD 3 DL+0.75*LR+0.75*WL2
56. SELFWEIGHT Y -1
57. JOINT LOAD
58. 145 FX 0.1 FY -8
59. 194 FX 0 FY -9 FZ 1.2
60. 146 FX 0 FY -10 FZ 1.75
61. ******************************************************
62. LOAD 4 DL+0.75*LR+0.75*WL3
63. SELFWEIGHT Y -1
64. JOINT LOAD
65. 145 FY -8 FZ -0.1
66. 194 FX 0 FY -9 FZ 1.2
67. 146 FX 0 FY -10 FZ 1.75
68. ******************************************************
69. LOAD 5 DL+0.75*LR+0.75*WL4
70. SELFWEIGHT Y -1
71. JOINT LOAD
72. 145 FY -8 FZ -0.1
73. 194 FX 0 FY -9 FZ 1.2
74. 146 FX 0 FY -10 FZ 1.75
75. ******************************************************
76. LOAD 6 0.6*DL+WL1
77. SELFWEIGHT Y -1
78. JOINT LOAD
79. 145 FX -0.1 FY -1.5
80. 194 FX 0 FY -2.8 FZ 0.6
81. 146 FX 0 FY -2.8 FZ 0.6
82. ******************************************************
83. LOAD 7 0.6*DL+WL2
84. SELFWEIGHT Y -1
85. JOINT LOAD
86. 145 FX 0.1 FY -1.5
87. 194 FX 0 FY -2.8 FZ 0.6
88. 146 FX 0 FY -2.8 FZ 0.6
89. ******************************************************
90. LOAD 8 0.6*DL+WL3
91. SELFWEIGHT Y -1
92. JOINT LOAD
93. 145 FY -1.5 FZ -0.1
94. 194 FX 0 FY -2.8 FZ 0.6
```

```
146 FX 0 FY -2.8 FZ 0.6
**************************************************
LOAD 9 0.6*DL+WL4
SELFWEIGHT Y -1
JOINT LOAD
145 FY -1.5 FZ 0.1
194 FX 0 FY -2.8 FZ 0.6
146 FX 0 FY -2.8 FZ 0.6
**********************************************************************************
LOAD 10 DL+0.75*LR+0.75*0.7*EL1
JOINT LOAD
145 FX -0.05 FY -8
194 FX 0 FY -9 FZ 1.2
146 FX 0 FY -10 FZ 1.75
SELFWEIGHT Y -1
**********************************************************
LOAD 11 DL+0.75*LR+0.75*0.7*EL2
SELFWEIGHT Y -1
JOINT LOAD
145 FX 0.05 FY -8
194 FX 0 FY -9 FZ 1.2
146 FX 0 FY -10 FZ 1.75
**********************************************************
LOAD 12 DL+0.75*LR+0.75*0.7*EL3
SELFWEIGHT Y -1
JOINT LOAD
145 FY -8 FZ -0.05
194 FX 0 FY -9 FZ 1.2
146 FX 0 FY -10 FZ 1.75
**********************************************************
LOAD 13 DL+0.75*LR+0.75*0.7*EL4
SELFWEIGHT Y -1
JOINT LOAD
145 FY -8 FZ 0.05
194 FX 0 FY -9 FZ 1.2
146 FX 0 FY -10 FZ 1.75
**********************************************************
LOAD 14 0.6*DL+0.7*EL1
SELFWEIGHT Y -1
JOINT LOAD
145 FX -0.1 FY -1.5
194 FX 0 FY -2.8 FZ 0.6
146 FX 0 FY -2.8 FZ 0.6
**********************************************************
LOAD 15 0.6*DL+0.7*EL2
SELFWEIGHT Y -1
JOINT LOAD
145 FX 0.1 FY -1.5
194 FX 0 FY -2.8 FZ 0.6
146 FX 0 FY -2.8 FZ 0.6
**********************************************************
LOAD 16 0.6*DL+0.7*EL3
SELFWEIGHT Y -1
JOINT LOAD
145 FY -1.5 FZ -0.1
194 FX 0 FY -2.8 FZ 0.6
```

```
151. 146 FX 0 FY -2.8 FZ 0.6
152. ************************************************
153. LOAD 17 0.6*DL+0.7*EL4
154. SELFWEIGHT Y -1
155. JOINT LOAD
156. 145 FY -1.5 FZ 0.1
157. 194 FX 0 FY -2.8 FZ 0.6
158. 146 FX 0 FY -2.8 FZ 0.6
159. ********************************************************************************
160. *****ULTIMATE STRENGTH DESIGN LOAD COMBS*************************************
161. LOAD 100 1.4*DL
162. SELFWEIGHT Y -1
163. JOINT LOAD
164. 145 FY -3.5
165. 194 FX 0 FY -7 FZ 1
166. 146 FX 0 FY -7 FZ 1.4
167. ************************************************
168. LOAD 101 1.4D+1.7(LL)
169. SELFWEIGHT Y -1
170. JOINT LOAD
171. 145 FY -15
172. 194 FX 0 FY -17 FZ 2
173. 146 FX 0 FY -18 FZ 3.1
174. ************************************************
175. LOAD 102 1.2D+1.6WL1+0.5L
176. SELFWEIGHT Y -1
177. JOINT LOAD
178. 145 FX -0.2 FY -6.5
179. 194 FX 0 FY -9.2 FZ 1.2
180. 146 FX 0 FY -9.2 FZ 1.7
181. ************************************************
182. LOAD 103 1.2D+1.6WL2+0.5L
183. SELFWEIGHT Y -1
184. JOINT LOAD
185. 145 FX 0.2 FY -6.5
186. 194 FX 0 FY -9.2 FZ 1.2
187. 146 FX 0 FY -9.2 FZ 1.7
188. ************************************************
189. LOAD 104 1.2D+1.6WL3+0.5L
190. SELFWEIGHT Y -1
191. JOINT LOAD
192. 145 FY -6.5 FZ -0.2
193. 194 FX 0 FY -9.2 FZ 1.2
194. 146 FX 0 FY -9.2 FZ 1.7
195. ************************************************8
196. LOAD 105 1.2D+1.6WL4+0.5L
197. SELFWEIGHT Y -1
198. JOINT LOAD
199. 145 FY -6.5 FZ 0.2
200. 194 FX 0 FY -9.2 FZ 1.2
201. 146 FX 0 FY -9.2 FZ 1.7
202. ********************************************************************************
203. LOAD 106 0.9D+1.6WL1
204. SELFWEIGHT Y -1
205. JOINT LOAD
206. 145 FX -0.2 FY -2.3
```

```
207. 194 FX 0 FY -4.2 FZ 1
208. 146 FX 0 FY -4.2 FZ 1
209. ************************************************************************
210. LOAD 107 0.9D+1.6WL2
211. SELFWEIGHT Y -1
212. JOINT LOAD
213. 145 FX 0.2 FY -2.3
214. 194 FX 0 FY -4.2 FZ 1
215. 146 FX 0 FY -4.2 FZ 1
216. ************************************************************************
217. LOAD 108 0.9D+1.6WL3
218. SELFWEIGHT Y -1
219. JOINT LOAD
220. 145 FY -2.3 FZ -0.2
221. 194 FX 0 FY -4.2 FZ 1
222. 146 FX 0 FY -4.2 FZ 1
223. ************************************************************************
224. LOAD 109 0.9D+1.6WL4
225. SELFWEIGHT Y -1
226. JOINT LOAD
227. 145 FY -2.3 FZ 0.2
228. 194 FX 0 FY -4.2 FZ 1
229. 146 FX 0 FY -4.2 FZ 1
230. ************************************************************************
231. LOAD 110 1.2D+1.0EL1+0.5L
232. SELFWEIGHT Y -1
233. JOINT LOAD
234. 145 FX -0.1 FY -6.5
235. 194 FX 0 FY -9.2 FZ 1.2
236. 146 FX 0 FY -9.2 FZ 1.7
237. ************************************************
238. LOAD 111 1.2D+1.0EL2+0.5L
239. SELFWEIGHT Y -1
240. JOINT LOAD
241. 145 FX 0.1 FY -6.5
242. 194 FX 0 FY -9.2 FZ 1.2
243. 146 FX 0 FY -9.2 FZ 1.7
244. ************************************************
245. LOAD 112 1.2D+1.0EL3+0.5L
246. SELFWEIGHT Y -1
247. JOINT LOAD
248. 145 FY -6.5 FZ -0.1
249. 194 FX 0 FY -9.2 FZ 1.2
250. 146 FX 0 FY -9.2 FZ 1.7
251. ************************************************8
252. LOAD 113 1.2D+1.0EL4+0.5L
253. SELFWEIGHT Y -1
254. JOINT LOAD
255. 145 FY -6.5 FZ 0.1
256. 194 FX 0 FY -9.2 FZ 1.2
257. 146 FX 0 FY -9.2 FZ 1.7
258. ************************************************************************
259. LOAD 114 0.9D+1.0EL1
260. SELFWEIGHT Y -1
261. JOINT LOAD
262. 145 FX -0.1 FY -2.2
```

```
263. 194 FX 0 FY -4.2 FZ 1
264. 146 FX 0 FY -4.2 FZ 1
265. ************************************************************************
266. LOAD 115 0.9D+1.0EL2
267. SELFWEIGHT Y -1
268. JOINT LOAD
269. 145 FX 0.1 FY -2.2
270. 194 FX 0 FY -4.2 FZ 1
271. 146 FX 0 FY -4.2 FZ 1
272. ************************************************************************
273. LOAD 116 0.9D+1.0EL3
274. SELFWEIGHT Y -1
275. JOINT LOAD
276. 145 FY -2.2 FZ -0.1
277. 194 FX 0 FY -4.2 FZ 1
278. 146 FX 0 FY -4.2 FZ 1
279. ************************************************************************
280. LOAD 117 0.9D+1.0EL4
281. SELFWEIGHT Y -1
282. JOINT LOAD
283. 145 FY -2.2 FZ 0.1
284. 194 FX 0 FY -4.2 FZ 1
285. 146 FX 0 FY -4.2 FZ 1
286. ************************************************************************
287. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**8**	**NUMBER OF MEMBERS**	**7**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 4/ 4/ 18 DOF
TOTAL PRIMARY LOAD CASES = 35, TOTAL DEGREES OF FREEDOM = 36
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 271986.8 MB

```
288. LOAD LIST 1 TO 17 100 TO 117
289. PRINT SUPPORT REACTION
```

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
151	1	4.83	23.31	-0.79	-4.24	1.22	-5.24
	2	4.28	20.94	-0.72	-3.85	1.10	-4.71
	3	4.21	20.88	-0.72	-3.85	1.10	-4.50
	4	4.24	20.91	-0.63	-3.47	1.13	-4.60
	5	4.24	20.91	-0.63	-3.47	1.13	-4.60
	6	2.46	11.34	-0.34	-1.75	0.48	-2.73
	7	2.39	11.28	-0.34	-1.75	0.48	-2.52
	8	2.42	11.31	-0.25	-1.36	0.51	-2.63
	9	2.42	11.31	-0.43	-2.13	0.46	-2.63
	10	4.26	20.93	-0.72	-3.85	1.10	-4.66
	11	4.22	20.90	-0.72	-3.85	1.10	-4.55
	12	4.24	20.91	-0.67	-3.66	1.12	-4.60
	13	4.24	20.91	-0.77	-4.04	1.09	-4.60
	14	2.46	11.34	-0.34	-1.75	0.48	-2.73
	15	2.39	11.28	-0.34	-1.75	0.48	-2.52
	16	2.42	11.31	-0.25	-1.36	0.51	-2.63
	17	2.42	11.31	-0.43	-2.13	0.46	-2.63
	100	3.65	15.41	-0.60	-3.17	0.90	-3.96
	101	6.59	31.91	-1.21	-6.54	1.89	-7.15
	102	4.37	19.57	-0.72	-3.82	1.09	-4.88
	103	4.23	19.46	-0.72	-3.82	1.09	-4.46
	104	4.30	19.51	-0.53	-3.05	1.14	-4.67
	105	4.30	19.51	-0.90	-4.59	1.04	-4.67
	106	2.90	12.87	-0.57	-2.91	0.80	-3.28
	107	2.76	12.75	-0.57	-2.91	0.80	-2.86
	108	2.83	12.81	-0.38	-2.14	0.85	-3.07
	109	2.83	12.81	-0.76	-3.68	0.75	-3.07
	110	4.34	19.54	-0.72	-3.82	1.09	-4.77
	111	4.27	19.48	-0.72	-3.82	1.09	-4.56
	112	4.30	19.51	-0.62	-3.44	1.12	-4.67
	113	4.30	19.51	-0.81	-4.20	1.07	-4.67
	114	2.87	12.74	-0.57	-2.91	0.80	-3.18
	115	2.80	12.68	-0.57	-2.91	0.80	-2.97
	116	2.83	12.71	-0.48	-2.52	0.83	-3.07
	117	2.83	12.71	-0.66	-3.29	0.78	-3.07
160	1	-4.83	25.91	-2.51	-10.61	-0.20	5.26
	2	-4.18	22.88	-2.23	-9.42	-0.21	4.48
	3	-4.31	22.94	-2.23	-9.42	-0.21	4.75
	4	-4.24	22.91	-2.22	-9.36	-0.19	4.62
	5	-4.24	22.91	-2.22	-9.36	-0.19	4.62
	6	-2.36	12.58	-0.86	-3.65	-0.18	2.50
	7	-2.49	12.64	-0.86	-3.65	-0.18	2.77
	8	-2.42	12.61	-0.85	-3.59	-0.15	2.64
	9	-2.42	12.61	-0.87	-3.72	-0.20	2.64
	10	-4.21	22.89	-2.23	-9.42	-0.21	4.55
	11	-4.27	22.92	-2.23	-9.42	-0.21	4.69

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	12	-4.24	22.91	-2.23	-9.39	-0.20	4.62
	13	-4.24	22.91	-2.23	-9.46	-0.22	4.62
	14	-2.36	12.58	-0.86	-3.65	-0.18	2.50
	15	-2.49	12.64	-0.86	-3.65	-0.18	2.77
	16	-2.42	12.61	-0.85	-3.59	-0.15	2.64
	17	-2.42	12.61	-0.87	-3.72	-0.20	2.64
	100	-3.65	18.91	-1.80	-7.63	-0.19	3.99
	101	-6.59	34.91	-3.89	-16.41	-0.31	7.17
	102	-4.17	22.15	-2.18	-9.23	-0.22	4.42
	103	-4.43	22.26	-2.18	-9.23	-0.22	4.96
	104	-4.30	22.21	-2.17	-9.10	-0.17	4.69
	105	-4.30	22.21	-2.20	-9.36	-0.28	4.69
	106	-2.70	14.65	-1.43	-6.09	-0.29	2.82
	107	-2.96	14.76	-1.43	-6.09	-0.29	3.36
	108	-2.83	14.71	-1.42	-5.96	-0.24	3.09
	109	-2.83	14.71	-1.44	-6.22	-0.34	3.09
	110	-4.24	22.18	-2.18	-9.23	-0.22	4.55
	111	-4.37	22.23	-2.18	-9.23	-0.22	4.82
	112	-4.30	22.21	-2.18	-9.16	-0.20	4.69
	113	-4.30	22.21	-2.19	-9.30	-0.25	4.69
	114	-2.77	14.68	-1.43	-6.09	-0.29	2.95
	115	-2.90	14.74	-1.43	-6.09	-0.29	3.22
	116	-2.83	14.71	-1.42	-6.03	-0.27	3.09
	117	-2.83	14.71	-1.44	-6.16	-0.32	3.09

********** END OF LATEST ANALYSIS RESULT **********

```
290. LOAD LIST 100 TO 117
291. START CONCRETE DESIGN
292. CODE ACI 1999
293. TRACK 0 MEMB 133 134 173 174 177
294. FC 3500 ALL
295. MAXMAIN 25 MEMB 133 134 173 174 177
296. CLB 0.075 MEMB 133 134 173 174 177
297. CLS 0.075 MEMB 133 134 173 174 177
298. DESIGN COLUMN 133 134 173 174 177
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 650.0 X 650.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4225.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.142	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 650.0 X 650.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4225.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.142	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==

COLUMN NO. 173 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 650.0 X 650.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4225.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.142	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==========

COLUMN NO. 174 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 650.0 X 650.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4225.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.142	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

==========

COLUMN NO. 177 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 34.3 MPA, SQRE SIZE - 650.0 X 650.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 4225.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.142	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

299. DESIGN BEAM 175 176

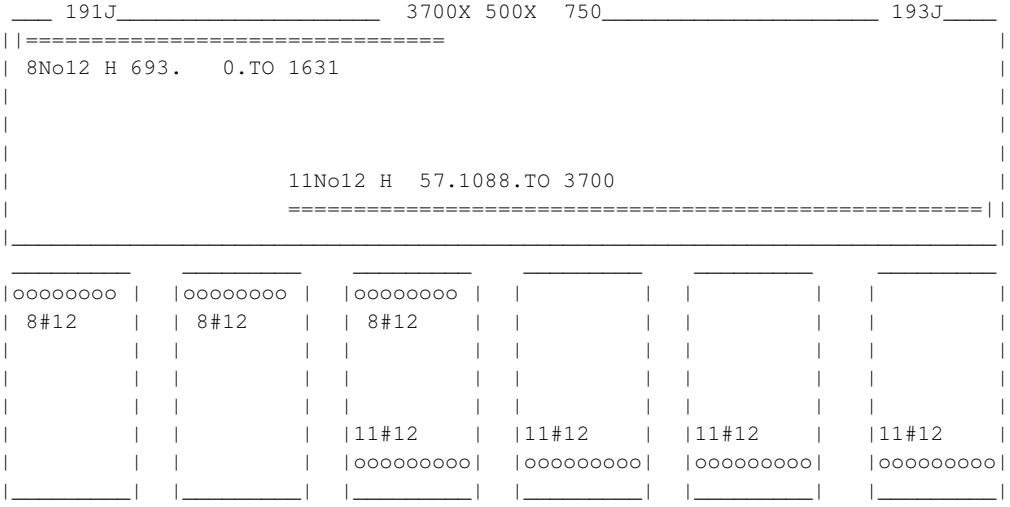
BEAM NO. 175 DESIGN RESULTS - FLEXURE PER CODE ACI 318-99

LEN - 3700. MM FY - 414. FC - 34. MPA, SIZE - 500. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	11 - 12MM	1088.	3700.	NO	YES
2	693.	8 - 12MM	0.	1631.	YES	NO

B E A M N O. 175 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 65.96 KNS Vc= 353.77 KNS Vs= 0.00 KNS
Tu= 10.63 KN-MET Tc= 23.3 KN-MET Ts= 0.0 KN-MET LOAD 100
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 45.40 KNS Vc= 328.94 KNS Vs= 0.00 KNS
Tu= 10.63 KN-MET Tc= 23.3 KN-MET Ts= 0.0 KN-MET LOAD 100
STIRRUPS ARE NOT REQUIRED.



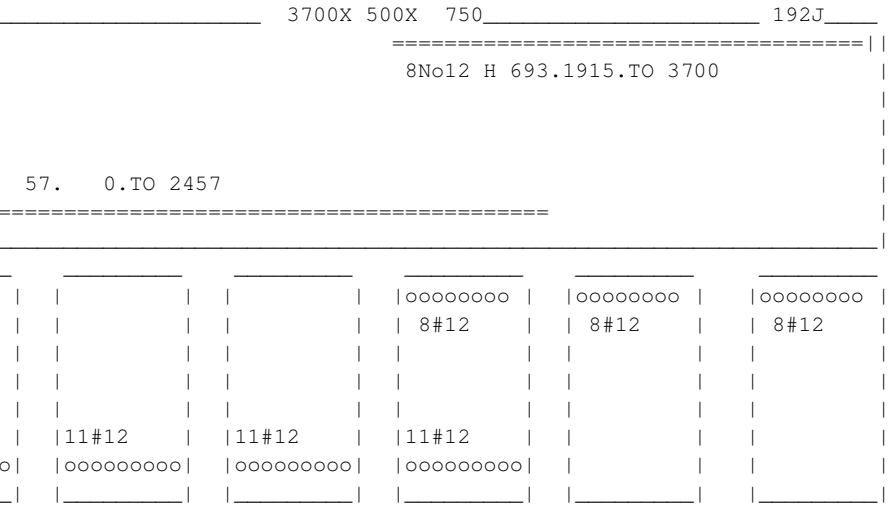
BEAM NO. 176 DESIGN RESULTS - FLEXURE PER CODE ACI 318-99

LEN - 3700. MM FY - 414. FC - 34. MPA, SIZE - 500. X 750. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	11 - 12MM	0.	2457.	YES	NO
2	693.	8 - 12MM	1915.	3700.	NO	YES

B E A M N O. 176 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 45.33 KNS Vc= 328.92 KNS Vs= 0.00 KNS
Tu= 0.83 KN-MET Tc= 23.3 KN-MET Ts= 0.0 KN-MET LOAD 100
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 65.89 KNS Vc= 353.97 KNS Vs= 0.00 KNS
Tu= 0.83 KN-MET Tc= 23.3 KN-MET Ts= 0.0 KN-MET LOAD 100
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

300. END CONCRETE DESIGN
301. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= NOV 18,2020 TIME= 12:48:13 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****